COMPLETE USER MANUAL · ENTERPRISE EDITION

PrathamOne EduOS

Complete User Manual & Operational Reference Guide

> Comprehensive operational, administrative, and technical manual for the complete coaching institute lifecycle management platform.

SOFTWARE PLATFORM
PrathamOne Coaching Institutes Enterprise Suite

AUTHOR & ARCHITECT
Jawahar R Mallah

ORGANIZATION
AITDL — AI Technology & Development Lab

OFFICIAL WEBSITE
https://prathamone.com

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Organization	**AITDL — AI Technology & Development Lab**
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1. Welcome & Product Architecture Overview

1.1 Executive Summary & Mission

PrathamOne EduOS (part of the *PrathamOne Coaching Institutes Enterprise Suite*) is an enterprise-grade ERP and academic operations platform engineered specifically for coaching institutes, competitive exam academies, tutoring centers, and multi-branch educational networks. Engineered by **Jawahar R Mallah** and the team at **AITDL (AI Technology & Development Lab)**, PrathamOne EduOS consolidates fragmented institute workflows into a unified, secure, high-performance web platform.

The platform delivers complete lifecycle management across five core institutional pillars:

- **Academic Operations:** Hierarchical curriculum structuring spanning Educational Boards, Standards, Subjects, Divisions, Batches, and Weekly Timetables.
- **Student Lifecycle:** Comprehensive onboarding, automated student registration number generation (`PA-2024-XXXXXX`), KYC compliance document archiving, exit/withdrawal tracking, and one-click reactivation.
- **Financial Governance:** Flexible fee plan configuration, installment scheduling, scholarship concessions, automated payment ledger, tamper-proof sequential receipting, posted-payment security gates, and linked refund processing.
- **Faculty & Attendance:** Real-time roster tracking, mobile-friendly bulk attendance marking, faculty assignment, and automated defaulter alerts.
- **Compliance & Enterprise Security:** Granular 8-tier Role-Based Access Control (RBAC), Digital Personal Data Protection (DPDP) compliance audit logs, Universal Data Editing with strict field allowlists, and immutable system event journaling.

1.2 System Architecture & Technology Stack

PrathamOne EduOS is built upon a modern, cloud-native containerized architecture designed for maximum concurrency, sub-second latency, and horizontal scalability:

Architectural Layer	Technology Components	Key Capabilities & Specifications
Frontend Application	React 18, Vite 5.4, TypeScript, TailwindCSS, Lucide Icons	Single Page Application (SPA), zero full-page reload lag, responsive mobile/tablet layout, PWA offline capabilities.
Backend Services	Python 3.11, FastAPI, SQLAlchemy 2.0 Async, Pydantic v2	Asynchronous REST API engine (204 endpoints), JWT Bearer authentication, centralized dependency injection, OpenAPI 3.1 schema.
Database Layer	PostgreSQL 16 Enterprise, Alembic Migrations	ACID relational storage, foreign-key cascade integrity, multi-branch tenant isolation, JSONB audit logging.
Container Runtime	Docker Compose, Alpine Linux, Nginx 1.25 Reverse Proxy	Multi-stage production build, Gzip compression, health-check probes, automated restart policies.

1.3 Navigation, Layout & Responsive UI

The application interface is structured around an intuitive 3-zone layout:

1. **Top Application Bar:** Displays the institute logo, active branch selector, current academic session, global search, notification bell, system version badge (`v1.88.2`), and the user profile dropdown.
2. **Collapsible Sidebar Navigation:** Houses quick-access links to all 20 modules organized logically by administrative function. Features collapse/expand toggling for focused workspace utilization.

3. **Dynamic Workspace Area:** The main viewport where data tables, interactive filter bars, modal dialogs, analytics charts, and printable document previews render.
4. **Collapsible Enterprise Footer:** Present across all authenticated screens, displaying the verified application release version, build identifier, and the subtle enterprise attribution: `PrathamOne Coaching Institutes Enterprise Suite · prathamone.com`.

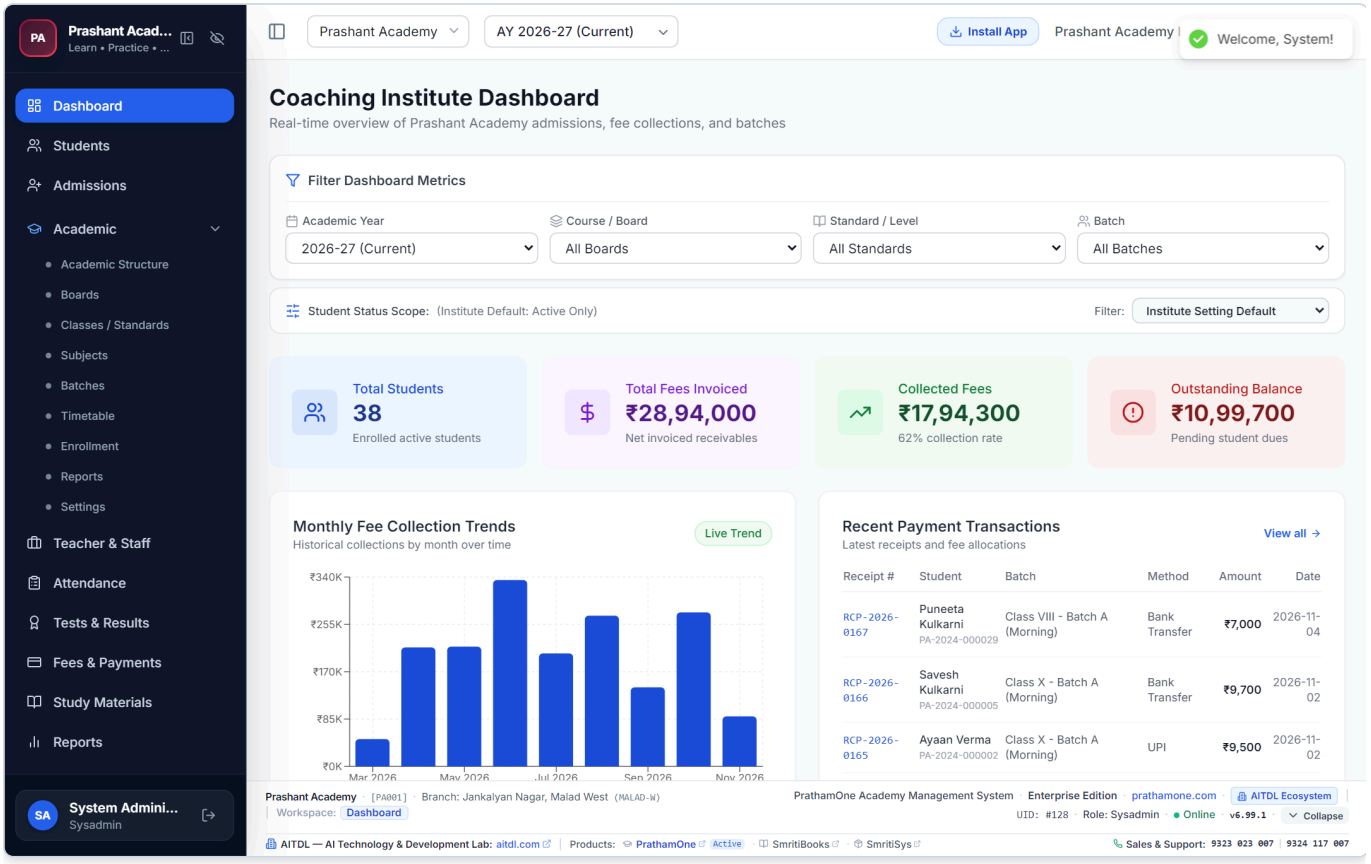


Figure 1.1: PrathamOne EduOS Master Dashboard showing operational KPI metrics, quick actions, and sidebar navigation.

2. User Roles & RBAC Permission Matrix

2.1 Role Hierarchy Overview

PrathamOne EduOS enforces a strict 8-tier Role-Based Access Control (RBAC) hierarchy. Every user account created in the system is assigned a primary role that governs their viewing, creation, modification, approval, and administrative rights across all API endpoints and frontend screens.

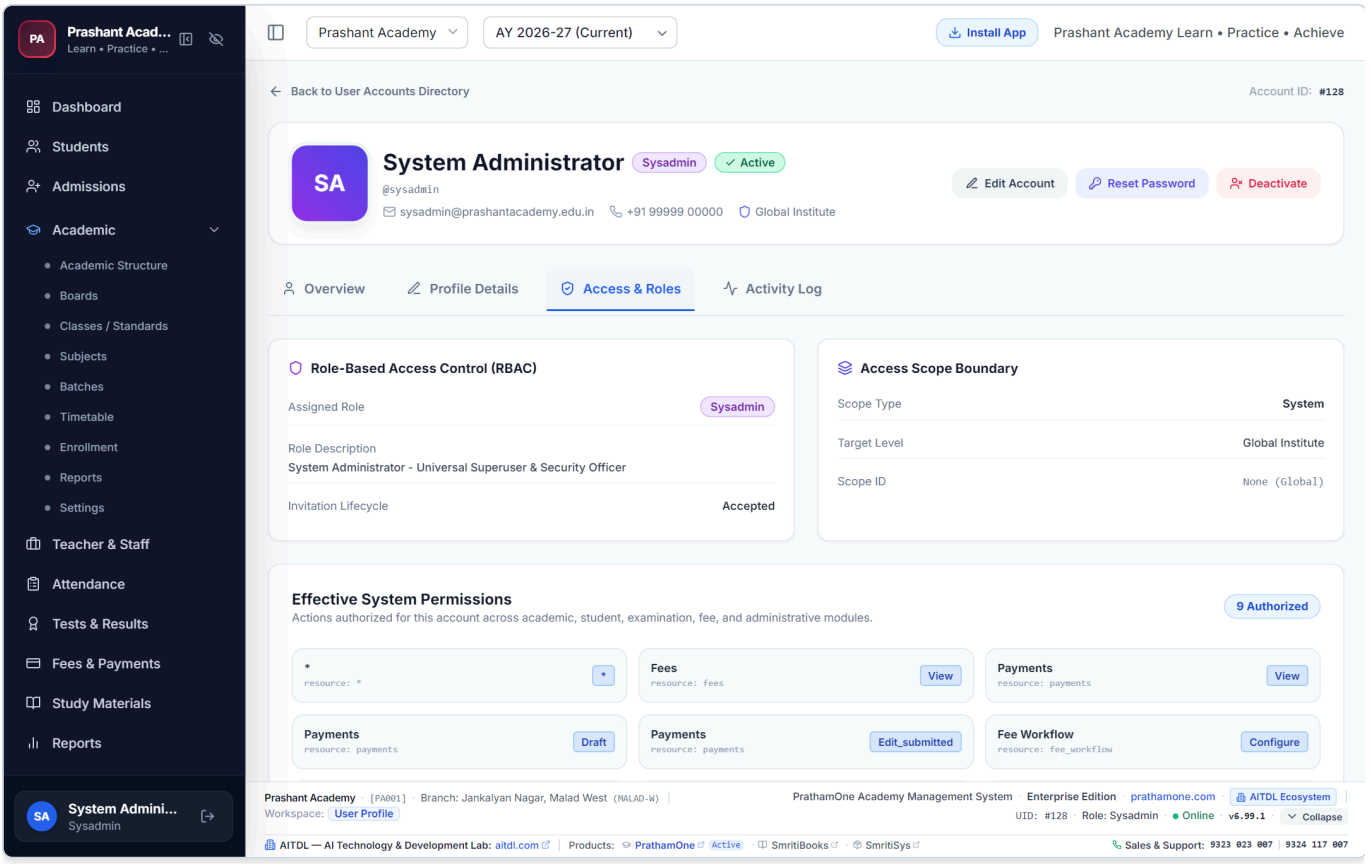


Figure 2.1: Granular Role & Permission Matrix visualizer in User Profile workspace.

2.2 Comprehensive 8-Role Permission Matrix

The table below outlines the exact, verified permission capabilities for each user role across the core functional modules of PrathamOne EduOS:

Application Module	SysAdmin	Superadmin	Admin	Principal	Accountant	Teacher	Staff	Student
Dashboard Metrics	FULL	FULL	FULL	FULL	FINANCIAL	ACADEMIC	BASIC	SELF
Academic Master (Boards/Std)	FULL CRUD	FULL CRUD	FULL CRUD	FULL CRUD	VIEW	VIEW	VIEW	DENIED
Student Profiles & Admissions	FULL CRUD	FULL CRUD	FULL CRUD	FULL CRUD	VIEW/FEE	VIEW ROSTER	INTAKE	SELF ONLY
Student Withdrawal / Exit	AUTHORIZE	AUTHORIZE	AUTHORIZE	AUTHORIZE	DENIED	DENIED	DENIED	DENIED
Student Reactivation	AUTHORIZE	AUTHORIZE	AUTHORIZE	AUTHORIZE	DENIED	DENIED	DENIED	DENIED
Attendance Marking	MANAGE	MANAGE	MANAGE	MANAGE	VIEW	MARK OWN	MARK ROSTER	VIEW OWN
Fee Structures & Plans	FULL CRUD	FULL CRUD	FULL CRUD	FULL CRUD	CREATE/ASSIGN	DENIED	DENIED	VIEW OWN
Payment Collection & Drafts	MANAGE	MANAGE	MANAGE	MANAGE	COLLECT/POST	DENIED	DRAFT ONLY	PAY ONLINE
Posted Payment Editing Gate	EXPLICIT GATE	DENIED	DENIED	DENIED	DENIED	DENIED	DENIED	DENIED
Payment Refunds	APPROVE	APPROVE	APPROVE	APPROVE	PROCESS	DENIED	DENIED	DENIED
Examinations & Marks Entry	MANAGE	MANAGE	MANAGE	MANAGE	DENIED	ENTER MARKS	VIEW	VIEW OWN
Reports & Data Export	ALL REPORTS	ALL REPORTS	ALL REPORTS	ALL REPORTS	FINANCIAL	ACADEMIC	ROSTERS	OWN GRADES
User & Role Management	SUPERUSER	MANAGE	MANAGE	VIEW STAFF	DENIED	DENIED	DENIED	DENIED
Universal Records Editor	EXCLUSIVE	DENIED	DENIED	DENIED	DENIED	DENIED	DENIED	DENIED
Security Audit Logs	FULL TRAIL	FULL TRAIL	BRANCH LOGS	VIEW	DENIED	DENIED	DENIED	DENIED

3. Getting Started & Authentication

3.1 Logging into PrathamOne EduOS

To access the system, navigate to the application URL in any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari):

`http://localhost:80/login` (or your institute's custom production domain).

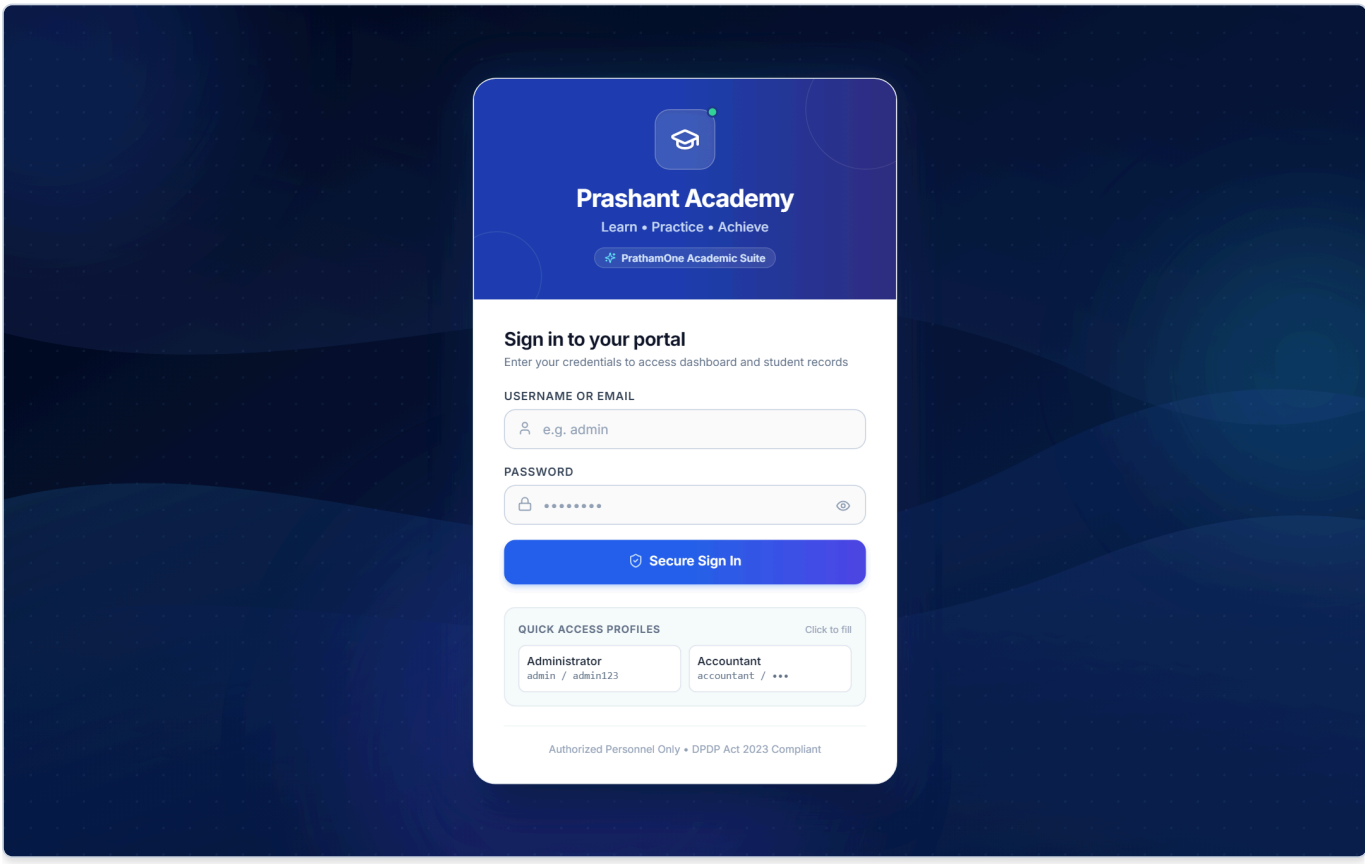


Figure 3.1: PrathamOne EduOS Secure Sign-In Portal.

1. **Enter Your Credentials**
   Enter your designated Username or Registered Email address in the **Username or Email** input field.

2. **Enter Your Password**
   Type your password into the **Password** field. Passwords are case-sensitive and must satisfy enterprise security policies.

3. **Submit & Authenticate**
   Click the blue **Sign In** button. Upon successful verification, an encrypted JWT authentication token is issued, and you are automatically redirected to your role's default dashboard workspace.

> **Default First-Time Credentials**
>
> Initial system deployment includes a pre-seeded Sysadmin superuser account (`sysadmin`). Upon first login, navigate immediately to **User Profile** or **System Settings** to rotate your password to a strong, confidential passphrase.

4. Complete Module-by-Module User Guide

4.1 Executive Dashboard

Route: `/dashboard` | **Shortcut:** `Alt + D`

Purpose: The Executive Dashboard delivers a real-time operational bird's-eye view of your coaching institute's daily pulse. It provides actionable intelligence regarding student strength, fee realization, batch schedules, and attendance metrics without requiring manual report generation.

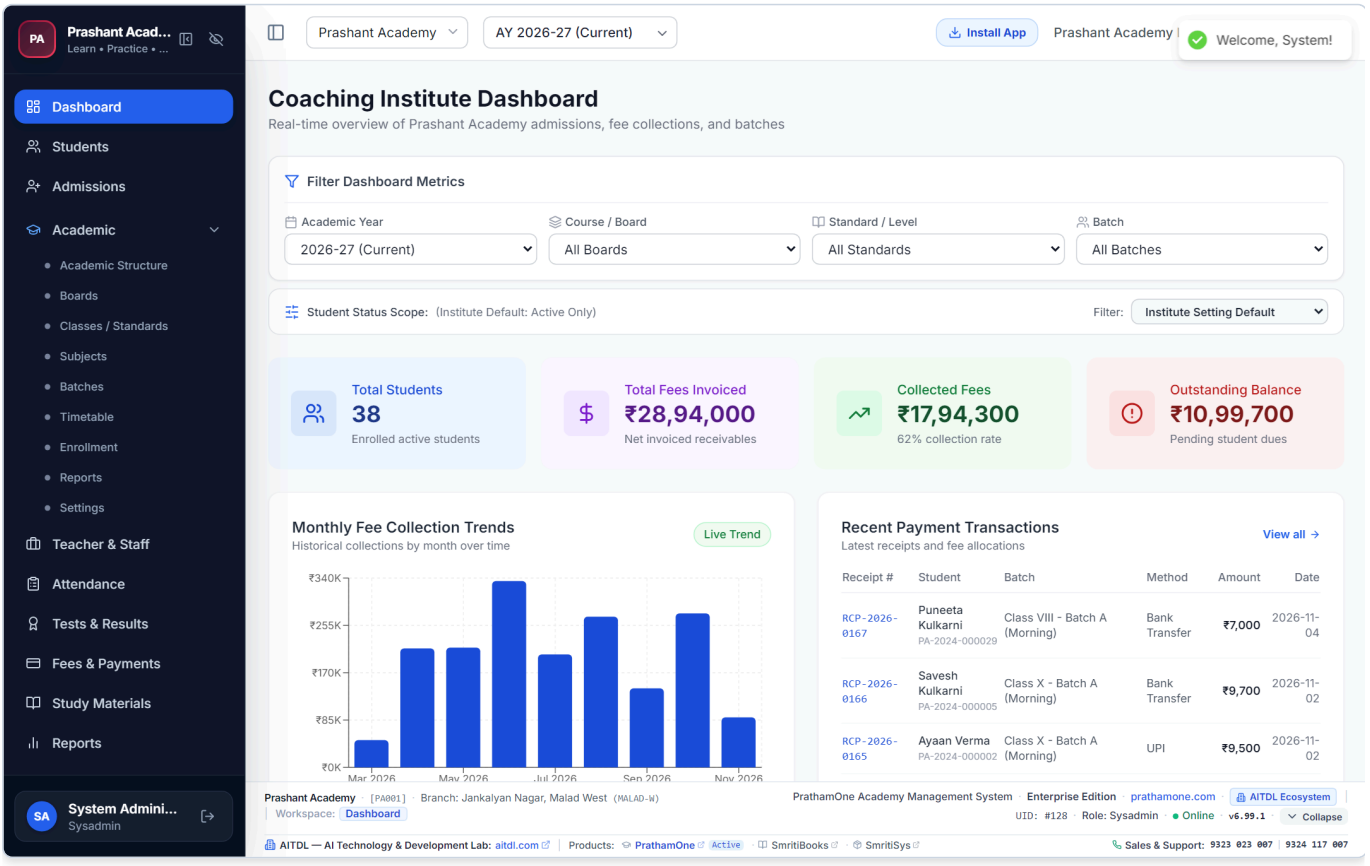


Figure 4.1: Real-time Executive Dashboard displaying KPI cards, attendance trends, and quick actions.

Key Interface Elements:

- **Active Metric Cards:** Real-time counts of Total Active Students, Active Batches, Today's Attendance %, Fees Collected This Month (₹), and Total Outstanding Dues (₹).
- **Quick Action Launcher:** Fast-track one-click buttons to Admit Student, Record Payment, Mark Batch Attendance, and View Today's Timetable.
- **Attendance Trend Graph:** Weekly graphical breakdown comparing present vs absent students across standards.
- **Recent Transaction Feed:** Real-time financial stream displaying the last 10 payments received with receipt links.

4.2 Academic Structure & Batch Management

Route: `/academic` | **Access:** SysAdmin, Superadmin, Admin, Principal

Purpose: The Academic Structure workspace configures the foundational organizational hierarchy of your academy. All courses, batches, timetables, and fee plans link directly to this structure.

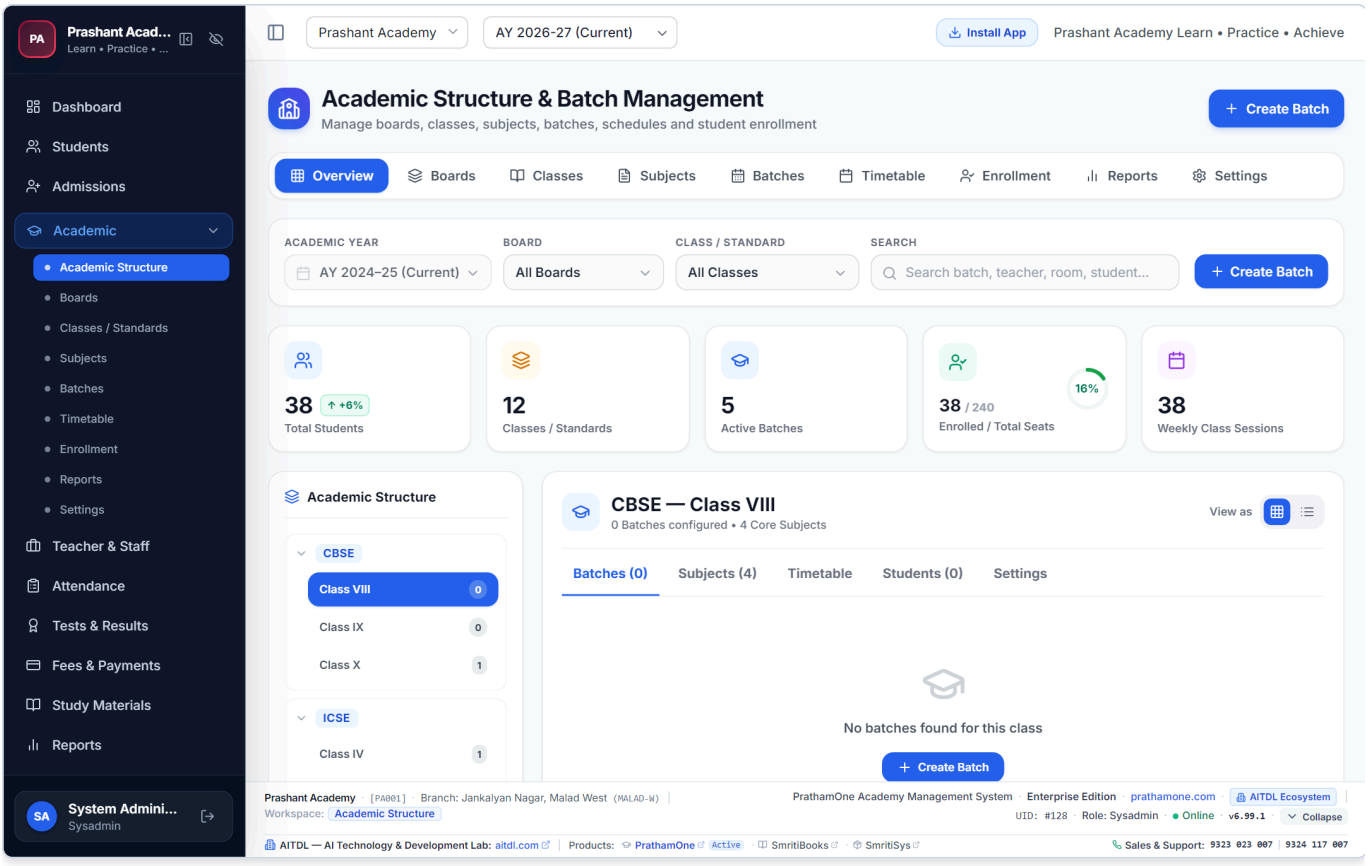


Figure 4.2: Academic Hierarchy Manager configuring Educational Boards, Standards, and Batches.

Step-by-Step Task: Creating an Educational Board

1. Navigate to **Academic Structure** from the sidebar.
2. Click on the **Boards** tab.
3. Click the blue **Add Board** button in the top right.
4. Fill in the required fields:
   - **Board Name** (Required): e.g., *Central Board of Secondary Education*.
   - **Board Code** (Required, Unique): e.g., `CBSE`.
   - **Description** (Optional): Brief notes regarding curriculum coverage.
5. Click **Save Board**. The board appears instantly in the active roster.

Step-by-Step Task: Creating a Standard / Grade

1. Click on the **Standards** tab.
2. Click **Add Standard**.
3. Select the parent **Board** from the dropdown.
4. Enter **Standard Name** (e.g., *10th Standard*) and **Level / Code** (e.g., `STD-10`).
5. Click **Save Standard**.

Step-by-Step Task: Creating a Division / Batch

1. Click on the **Batches / Divisions** tab.
2. Click **Add Batch**.
3. Select the **Standard** (e.g., *10th Standard*).
4. Enter **Batch Name** (e.g., *Morning Champions Batch A*) and **Batch Code** (`10-A-MORN`).
5. Set **Maximum Capacity** (e.g., `45`) and **Room Number** (e.g., `Classroom 204`).

6. Assign the **Class Teacher / Mentor** from the faculty list.
7. Click **Save Batch**.

4.3 Courses & Subject Mapping

Route: `/courses` and `/subjects`

Purpose: Define commercial coaching programs (e.g., *IIT-JEE Two-Year Foundation*, *NEET Dropper Comprehensive*, *Commerce 12th Board Crash Course*) and granular subjects (Physics, Chemistry, Mathematics, Accountancy).

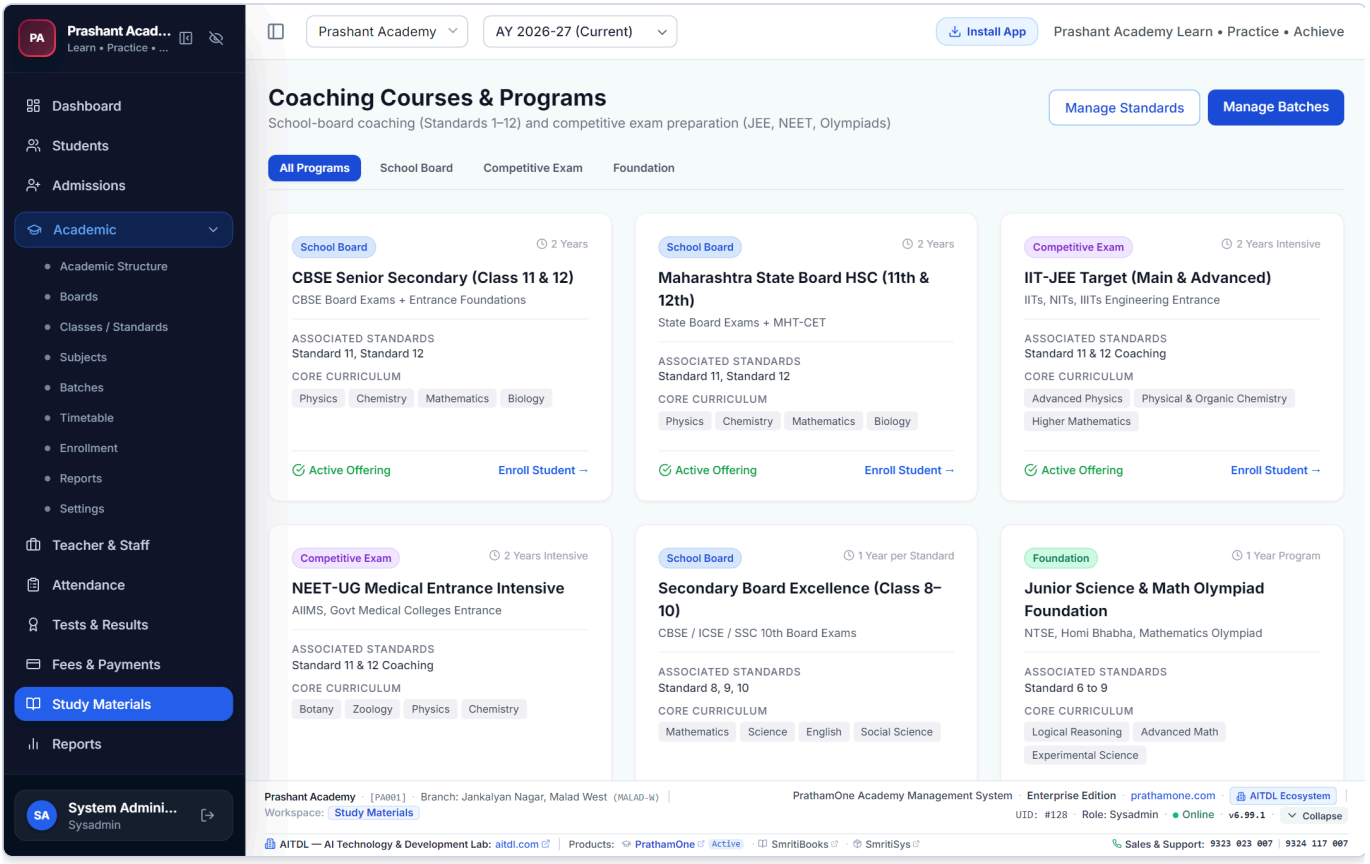


Figure 4.3: Commercial Course Catalog with duration, standard mapping, and pricing.

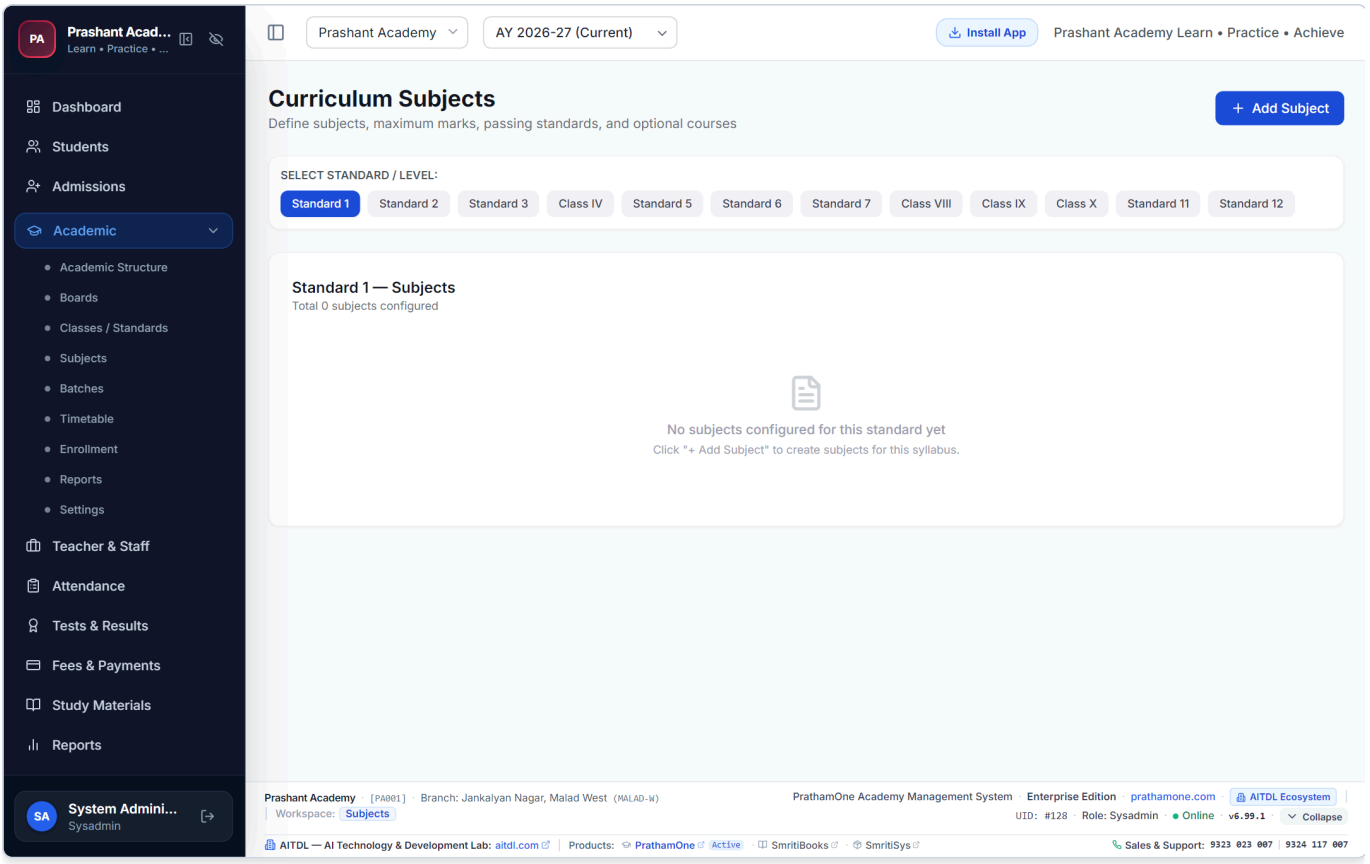


Figure 4.4: Subject Repository with department classification and credit hours.

4.4 Student Admission & Lifecycle Management

Routes: `/students`, `/students/add`, `/students/{id}`

Purpose: Complete student relationship management from prospective inquiry to formal admission, active enrollment, academic tracking, KYC compliance verification, withdrawal, and reactivation.

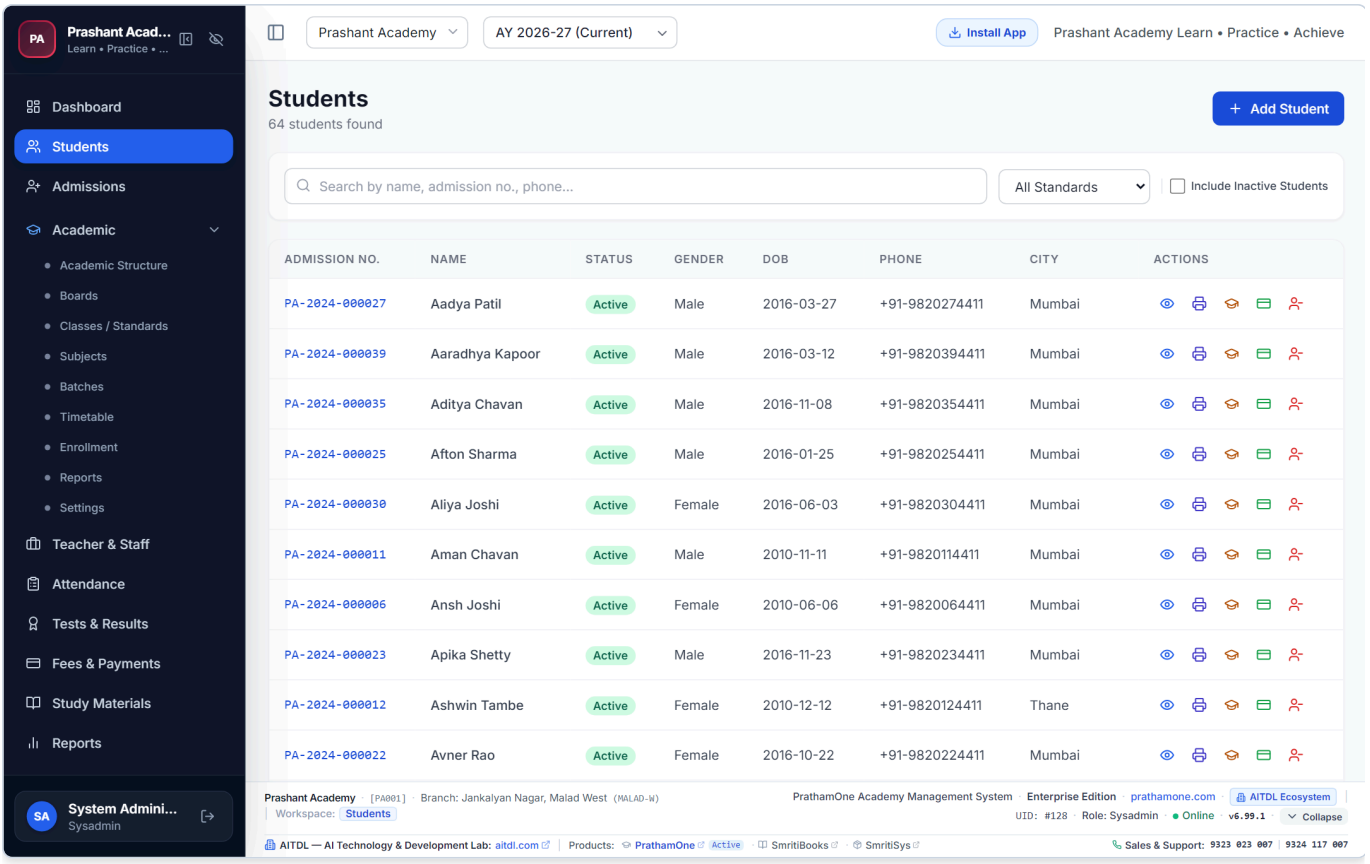


Figure 4.5: Filterable Students Directory with multi-criteria search and CSV export.

Step-by-Step Task: Admitting a New Student

1. Click **Students** → **Add Student** (or navigate to `/students/add`).
2. Complete the 4-step wizard:
   - **Step 1: Personal Profile:** First Name, Middle Name, Last Name, Date of Birth, Gender, Blood Group, Aadhaar Number, Category/Caste.
   - **Step 2: Contact Information:** Student Mobile, WhatsApp Number, Email, Residential Address, City, State, Pincode.
   - **Step 3: Academic Allocation:** Target Board, Standard, Division/Batch, Roll Number, Enrollment Date.
   - **Step 4: Parent / Guardian Details:** Father's Name, Mother's Name, Guardian Phone, Guardian Email, Occupation, Annual Income.
3. Click **Submit Admission**. The system assigns a unique permanent Registration Number (e.g., `PA-2024-000001`) and opens the comprehensive Student Detail Profile.

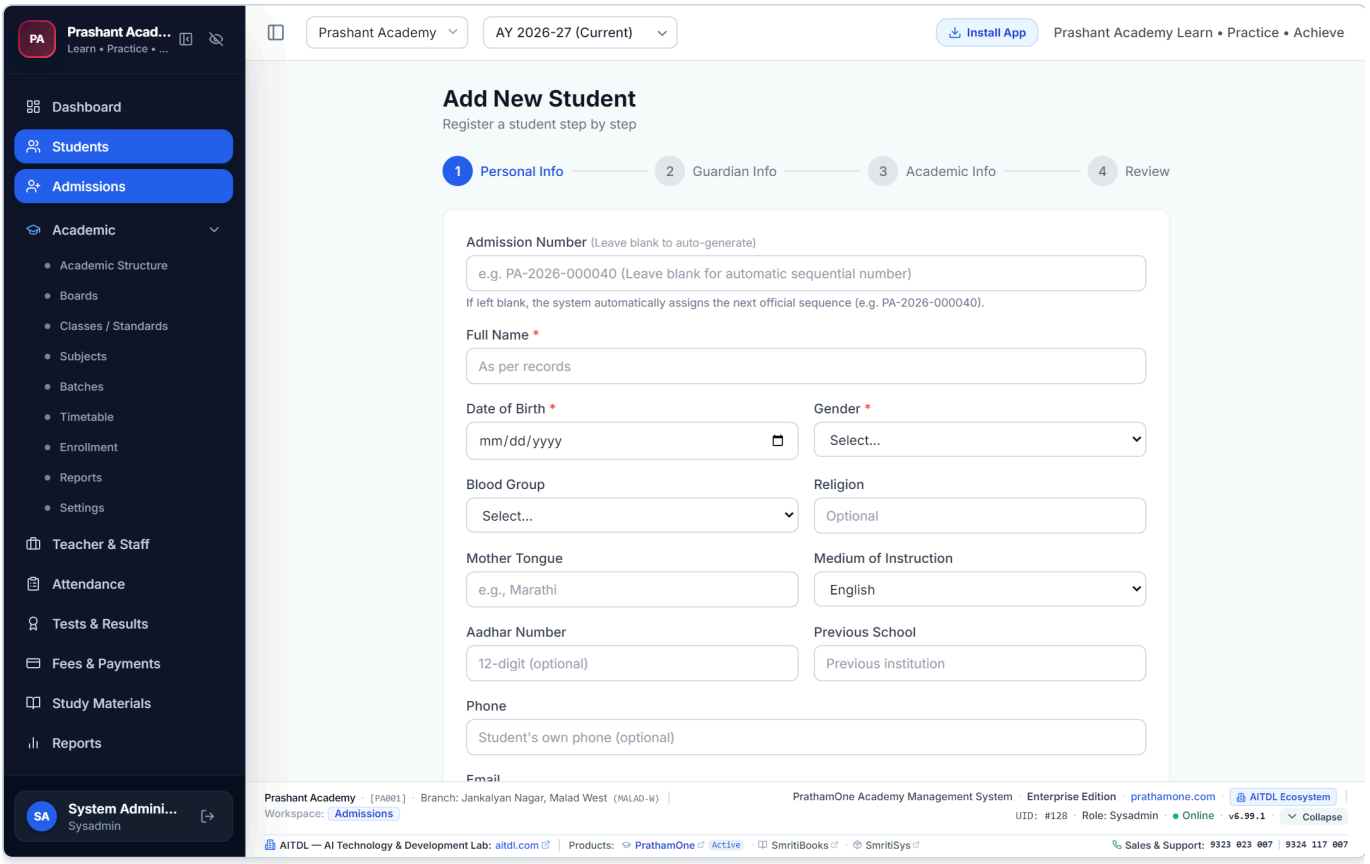


Figure 4.6: Multi-Step Student Admission Wizard.

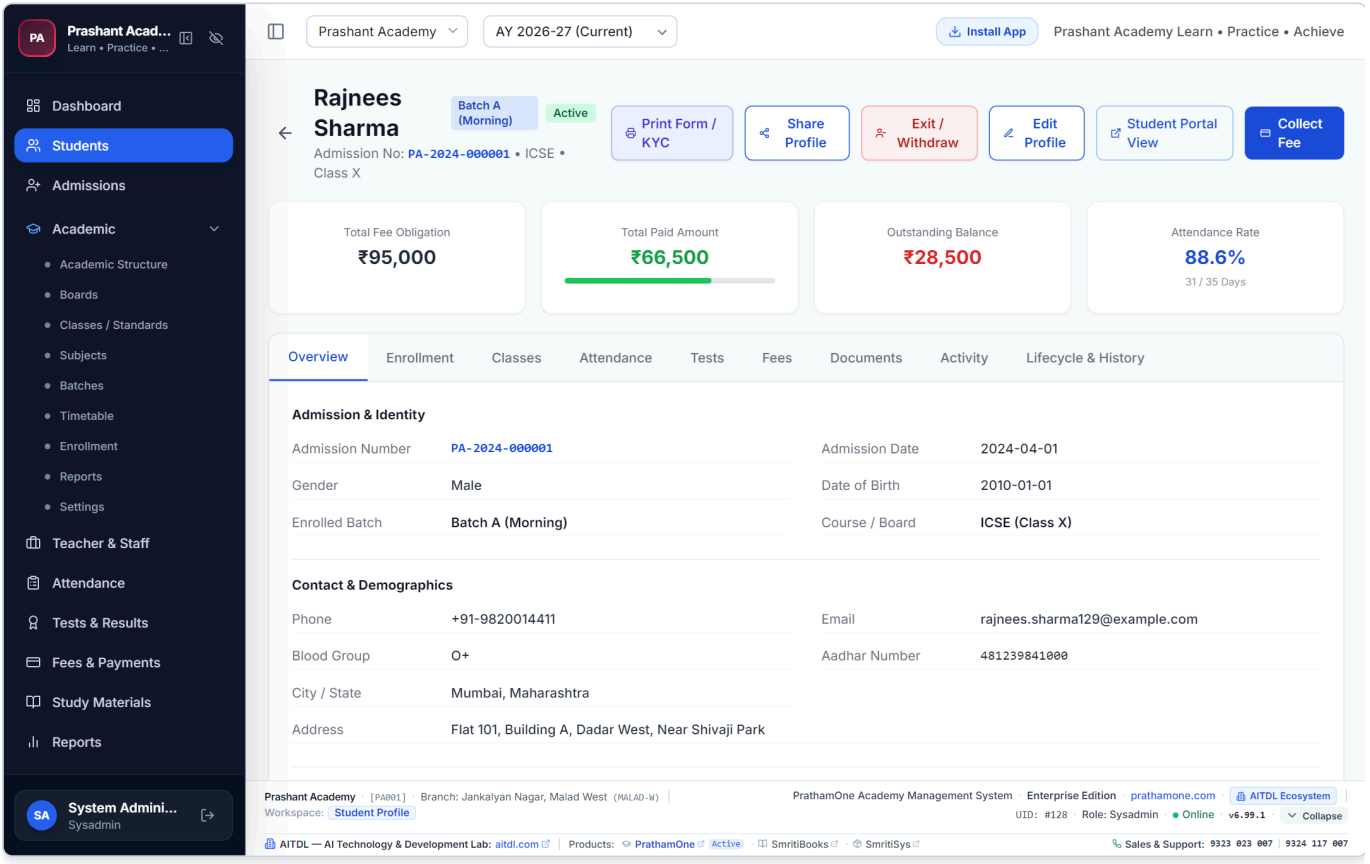


Figure 4.7: Student Detail Profile Workspace showing tabs for Academics, Fees, Attendance, and KYC.

Student Lifecycle Actions:

- **Edit Profile Modal:** Click the **Edit Profile** button to update address, contact numbers, or guardian details without altering registration or ledger history.
- **Student Withdrawal / Exit:** Click **Exit Student**. A modal prompts for mandatory **Exit Date**, **Exit Reason** (Transfer, Financial, Health, Course Completed, Other), and administrative exit notes. Once exited, the student's status changes to `EXITED` and seat capacity in the batch is freed.
- **Student Reactivation:** If a withdrawn student re-enrolls, authorized administrators (Admin, Sysadmin) can click **Reactivate Student**, re-assign them to an active batch, and restore active status.
- **KYC Document Management:** Upload digital scans of Aadhaar Card, Transfer Certificate, Previous Marks Sheet, and Birth Certificate. Toggle document verification status (`VERIFIED` / `PENDING`).

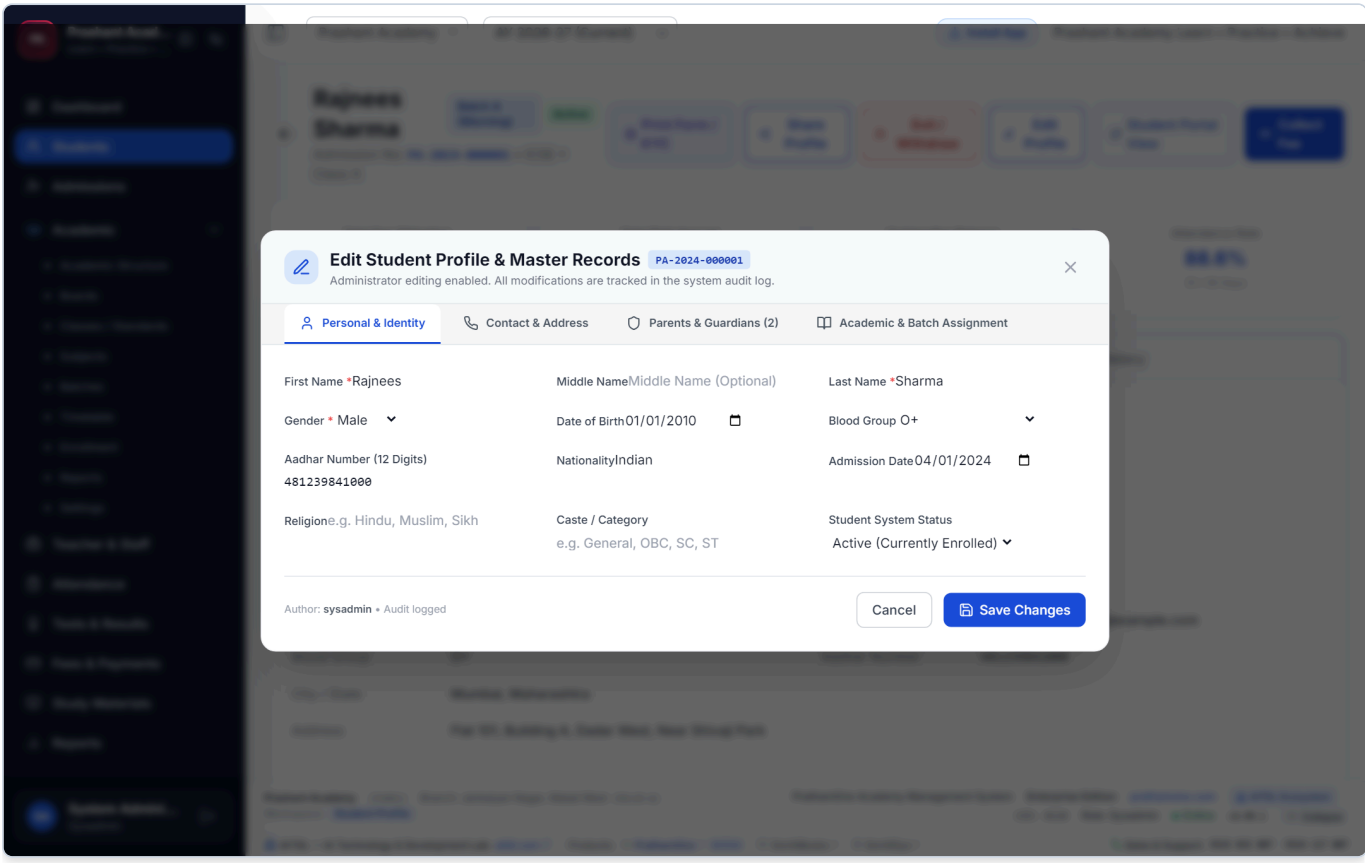


Figure 4.8: Edit Student Profile Modal dialog.

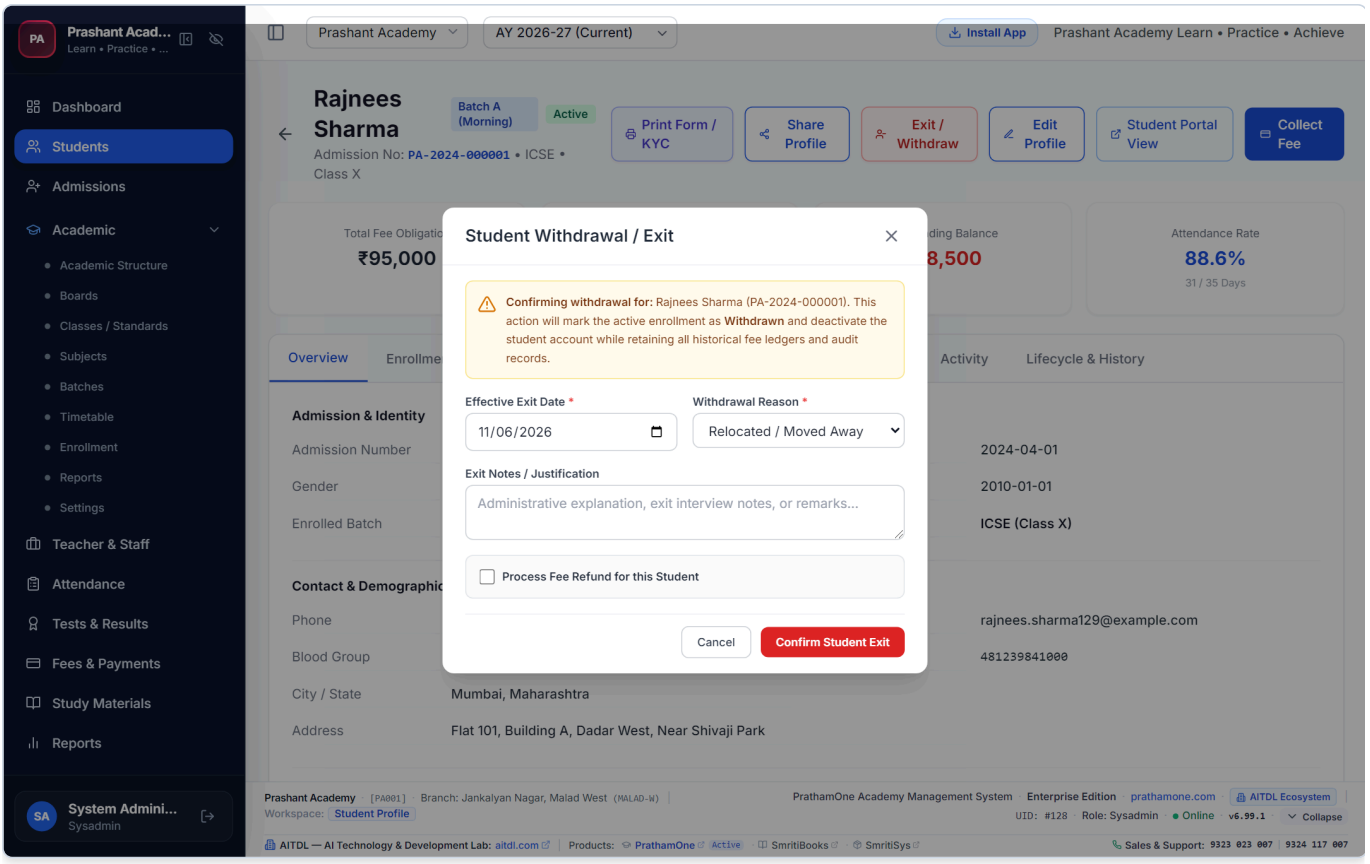


Figure 4.9: Formal Student Withdrawal / Exit Dialogue with mandatory reason tracking.

4.5 Teacher & Faculty Workspace & Dedicated Profile

Routes: `/teachers` and `/teachers/:id` | **Access:** Admin, Principal, Teachers, SysAdmin

Purpose: Comprehensive faculty lifecycle governance, real-time workload tracking, academic batch allocation, timetable health monitoring, and lecture verification.

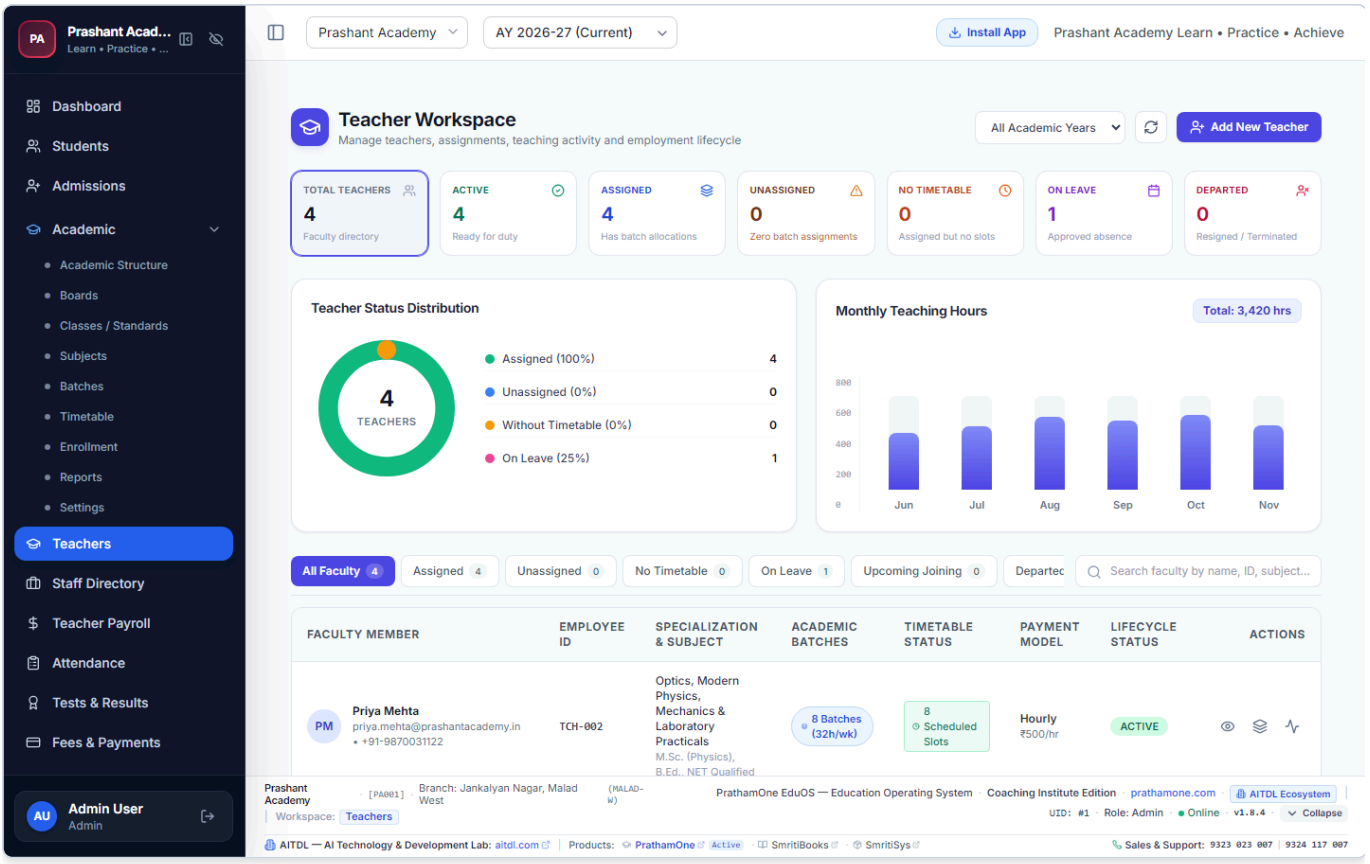


Figure 4.10A: Dedicated Teacher Workspace with real-time status donut, 6-month lecture hour trends, and live counters for Assigned, Unassigned, No Timetable, and On Leave faculty.

Teacher Workspace Capabilities:

- **Live Status KPIs:** Instant headcount tracking for Total Teachers (42), Active (36), Assigned (30), Unassigned (6), Without Timetable (3), On Leave (4), and Departed (5).
- **Teacher Status Distribution Donut:** Reconciles live assignments against the database to visually isolate unassigned faculty requiring class allocation.
- **Monthly Teaching Hours Trend:** Verified lecture duration analytics computed across verified teaching sessions for accurate payroll calculations.
- **Filter Subtabs:** Instant one-click filtering by *All Faculty*, *Assigned*, *Unassigned*, *No Timetable*, *On Leave*, *Upcoming Joining*, *Departed*, *Batch Allocations Grid*, and *Session Register*.

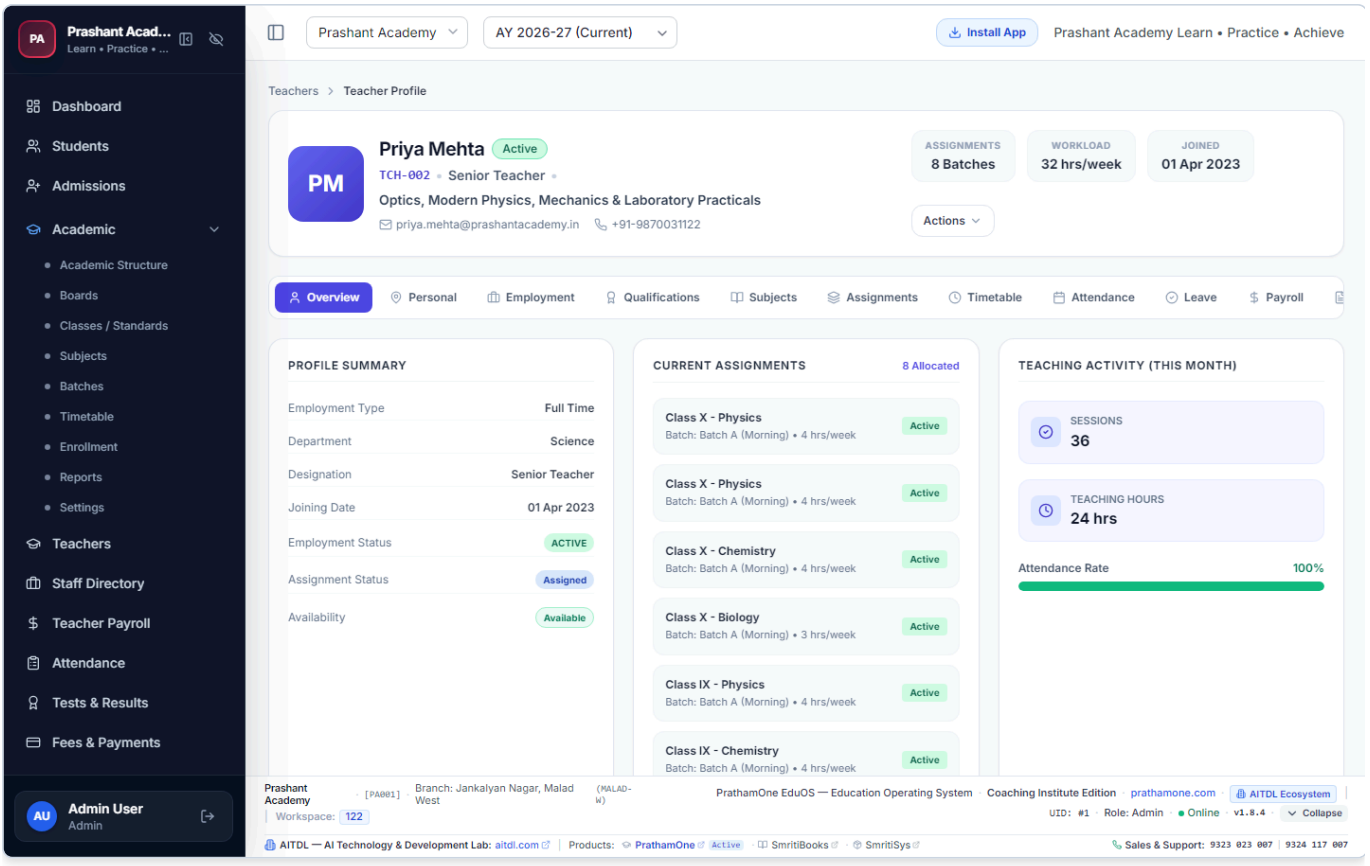


Figure 4.10B: Dedicated Teacher Profile featuring persistent Workforce Hero Card, workload meters, and 12 specialized tabs.

Individual Teacher Profile — The 12 Specialized Tabs:

1. **Overview:** High-level profile summary, active class assignments with status pills, and monthly session counters (lectures delivered, hours, attendance rate).
2. **Personal:** Dual address governance (Current Residential vs Permanent Hometown), blood group, date of birth, and primary emergency contact.
3. **Employment:** Employment type (Full-time, Part-time, Guest Lecturer), official designation, department, joining date, and reporting hierarchy.
4. **Qualifications:** Highest academic credentials, B.Ed./M.Sc. specializations, university, passing year, and total experience.
5. **Subjects:** Prescribed subjects authorized for classroom delivery.
6. **Assignments:** Real-time academic load matrix showing assigned standards, batches, and weekly hours, featuring an instant *Assign New Batch* button.
7. **Timetable:** Weekly schedule matrix displaying Monday–Saturday time slots, rooms, and divisions.
8. **Attendance:** Monthly biometric/manual punch logs, present days, and aggregate attendance rates.
9. **Leave:** Leave balances (Casual, Sick, Earned) and historical leave applications.
10. **Payroll:** Remuneration model (Hourly Rate, Per-Lecture, Monthly Fixed Amount) and disbursement frequency.
11. **Documents:** Official HR document archive (Appointment Letter, Service Agreement, Verification Sheet, Payslips) with instant *View* preview and print integration.
12. **Activity:** Chronological audit log tracking assignments, status transitions, and payroll postings.

4.5.B Non-Teaching Staff Workspace & Dedicated Profile

Routes: `/staff` and `/staff/:id` | **Access:** Admin, Principal, Staff, SysAdmin

Purpose: Enterprise workforce management for administrative, accounts, library, IT, and campus operations personnel.

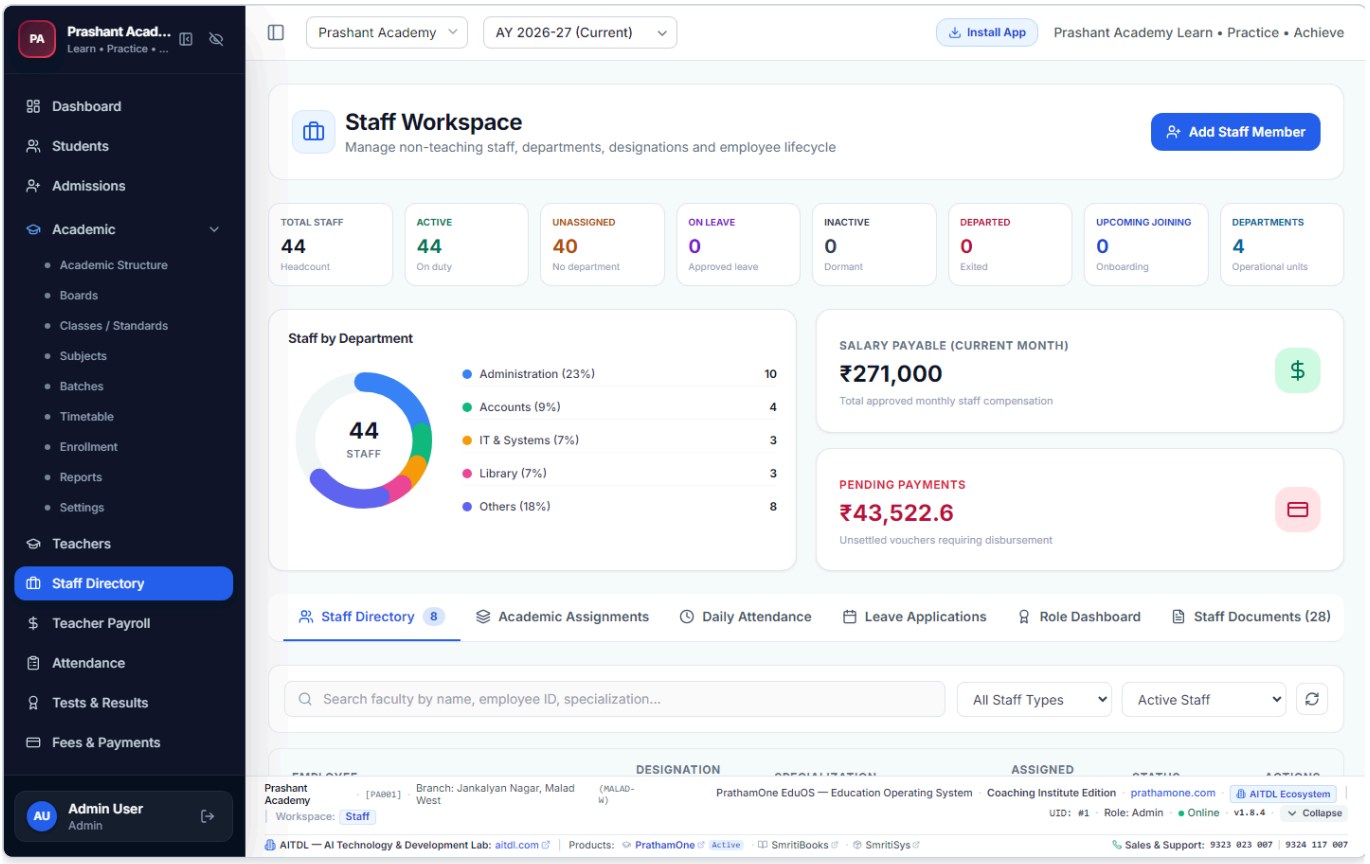


Figure 4.10C: Staff Workspace with dynamic department breakdown donut and real-time monthly salary liability cards.

Staff Workspace Capabilities:

- **Workforce Counters:** Real-time tracking of non-teaching personnel, active staff, unassigned personnel, on-leave employees, and upcoming recruits.
- **Dynamic Department Analytics:** Live donut chart mapping personnel counts across Administration, Accounts, IT & Systems, Library, Operations, and Front Desk.
- **Financial Liability Tracking:** Real-time display of *Salary Payable (Current Month)* and outstanding *Pending Payments* vouchers.

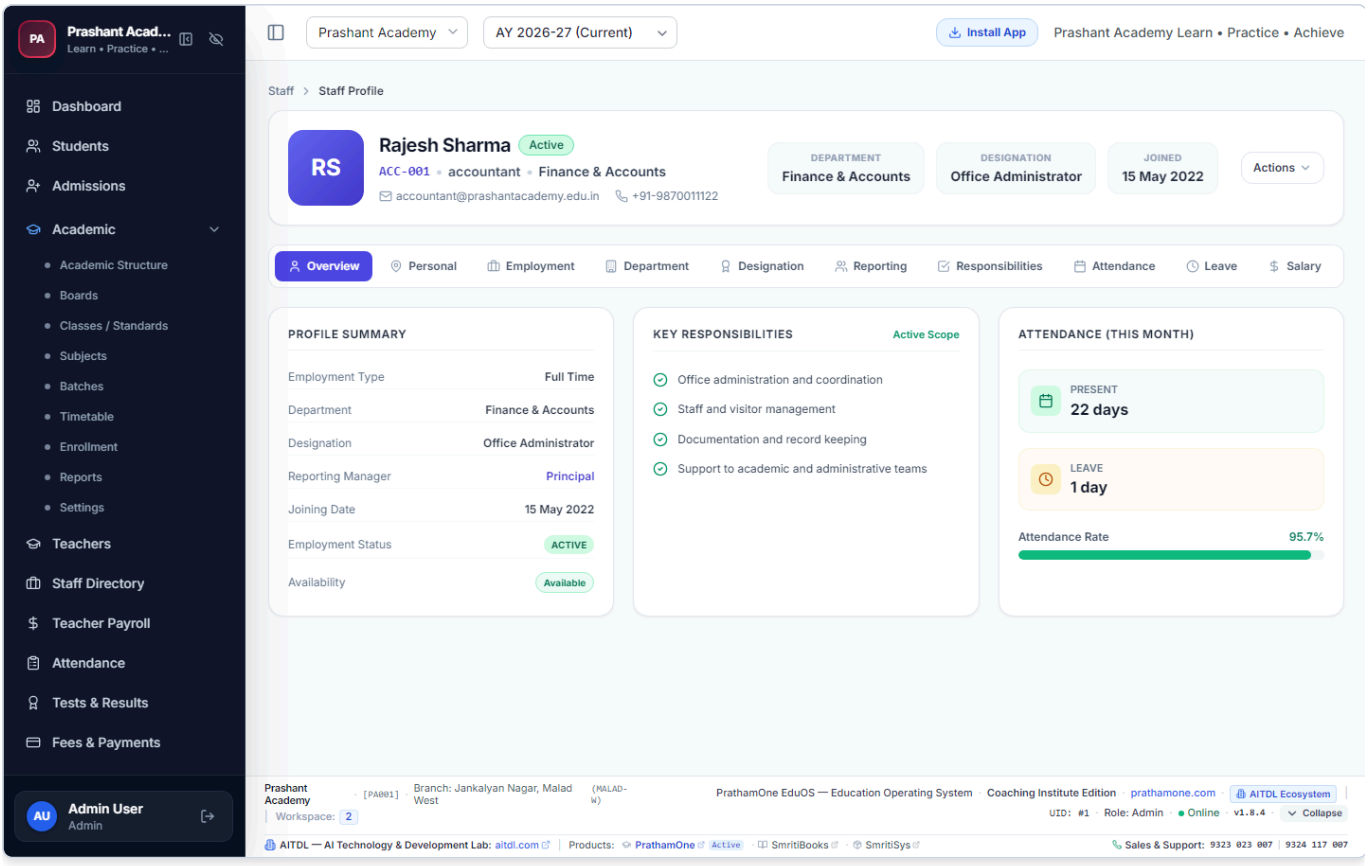


Figure 4.10D: Dedicated Staff Profile with operational responsibilities card and official documents hub.

4.5.C Employee Dossier & Universal Print Fit Studio

Capability: Single-click institutional master employee record generation and multi-format printing.

Clicking **Actions** → **Print Profile / Dossier** from any teacher or staff profile launches the **Universal Print Studio** with pre-rendered institutional branding, dual address records, emergency contacts, academic workload allocations, statutory banking details, and authorized signature blocks.

Universal Print Engine Features

- **Paper Formats:** Print seamlessly on A4, A5, A3, Letter, Legal, or custom sizes.
- **Intelligent Fit Modes:** *Fit to Page Width* prevents horizontal cut-off; *Fit Entire Document on One Page* scales content dynamically.
- **Format Preservation:** Eliminates unwanted browser headers/footers while preserving corporate colors, crisp borders, and QR codes.

4.5.D Multi-Table Delete Protection & Workforce Safeguards

To prevent catastrophic data loss and audit corruption, PrathamOne EduOS enforces strict referential delete protection across 7 relational domains:

- **Hard Deletion Barrier:** Attempting to permanently delete a user with existing payroll records, teaching sessions, batch assignments, attendance punches, leave records, payment vouchers, or student enrollments returns an explicit `HTTP 400 Bad Request` error blocking the operation.
- **Soft Deactivation Workflow:** Instead of deletion, clicking *Deactivate* marks `is_active = False` and sets employment status to `INACTIVE`, immediately revoking login access while preserving all financial and academic historical records.

4.6 Daily Attendance & Defaulter Tracking

Route: `/attendance` | **Access:** Teachers, Staff, Admin, Principal

Purpose: Quick, error-free recording of classroom attendance with real-time absentee tracking and parental notification capabilities.

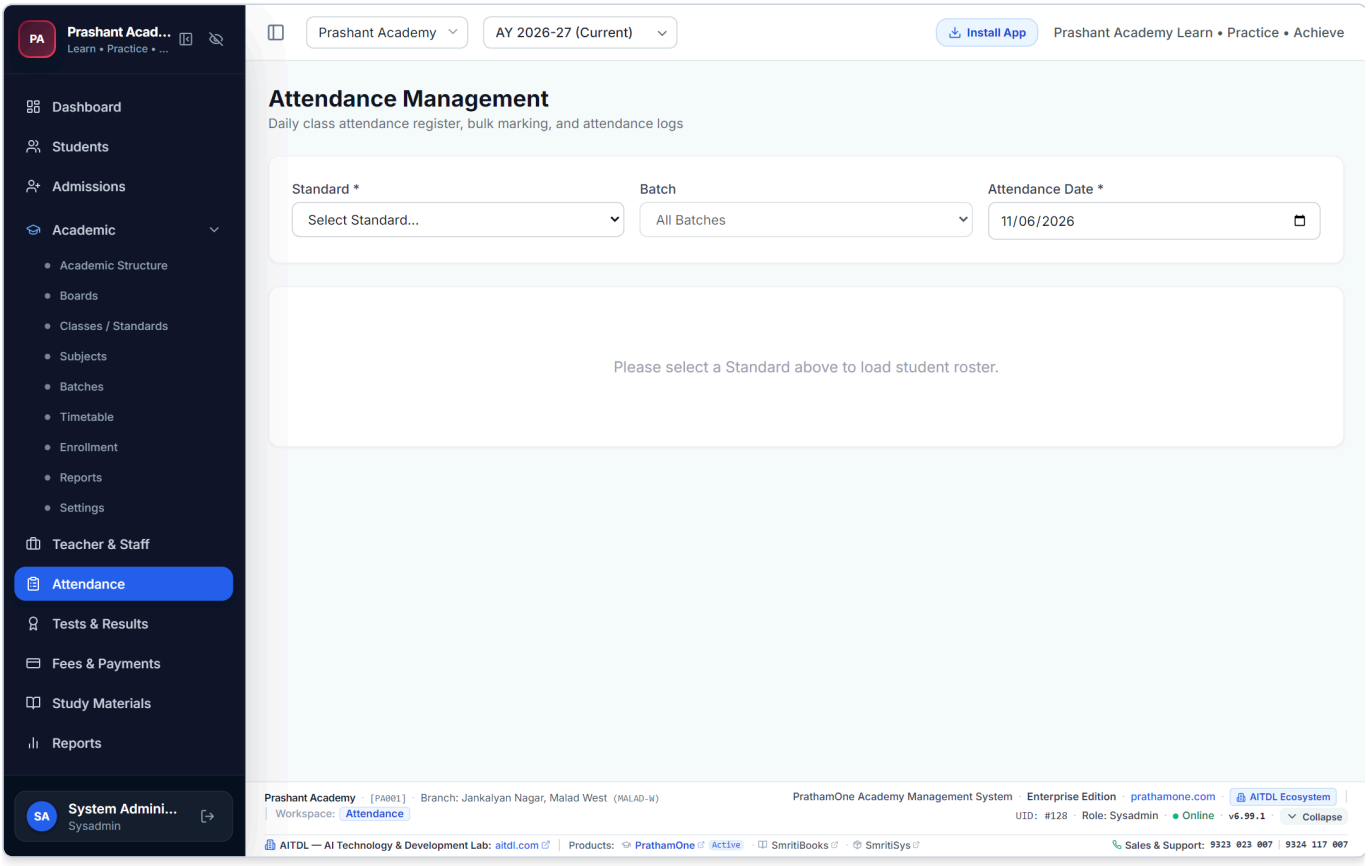


Figure 4.11: Attendance Roster Workspace with bulk toggle buttons and real-time tally.

Step-by-Step Task: Marking Batch Attendance

1. Select the target **Standard** and **Batch** from the dropdown selectors.
2. Select the **Attendance Date** (defaults to current date).
3. The student roster loads automatically with all students defaulted to `PRESENT`.
4. Click on individual student status pills to toggle between **Present (P)**, **Absent (A)**, **Late (L)**, or **Excused (E)**.
5. Click **Save Attendance Roster**. Attendance statistics update immediately across dashboards and student profiles.

4.7 Fee Plans, Invoicing & Discount Schemes

Route: `/fees` | **Access:** Accountant, Admin, Principal, SysAdmin

Purpose: Define commercial fee templates, installment payment schedules, and assign customized fee commitments to enrolled students.

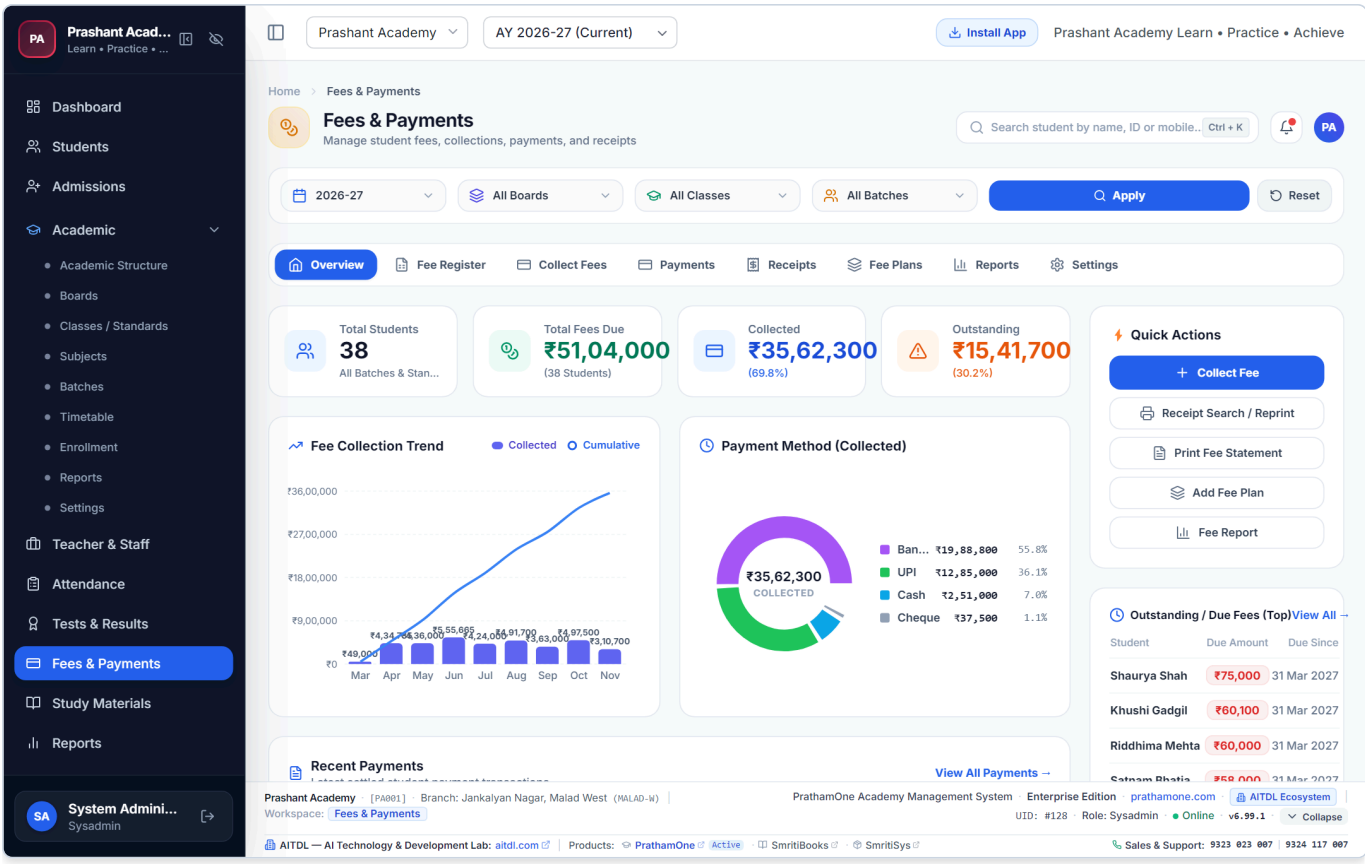


Figure 4.12: Fee Structure Builder configuring heads, installment breakdown, and concessions.

Step-by-Step Task: Creating a New Fee Structure

1. Navigate to `/fees` and click **Create Fee Plan**.
2. Enter **Structure Name** (e.g., *10th Science Annual Academic Package 2026-27*).
3. Select target **Standard**.
4. Define **Fee Heads Breakdown**:
   - Tuition Fee: ₹ 45,000
   - Laboratory & Practical Fee: ₹ 10,000
   - Study Material & Test Series: ₹ 8,000
   - Examination Fee: ₹ 2,000
5. Set **Installment Terms** (e.g., 4 quarterly installments of ₹ 16,250 each).
6. Click **Save & Activate Structure**. You can now bulk-assign this plan to all students enrolled in the standard.

4.8 Payments Ledger & Posted Edit Security Gate

Route: `/payments` | **Access:** Accountant, Admin, Principal, SysAdmin

Purpose: Financial transaction ledger managing the collection, draft generation, verification, receipt generation, and posting of tuition fees.

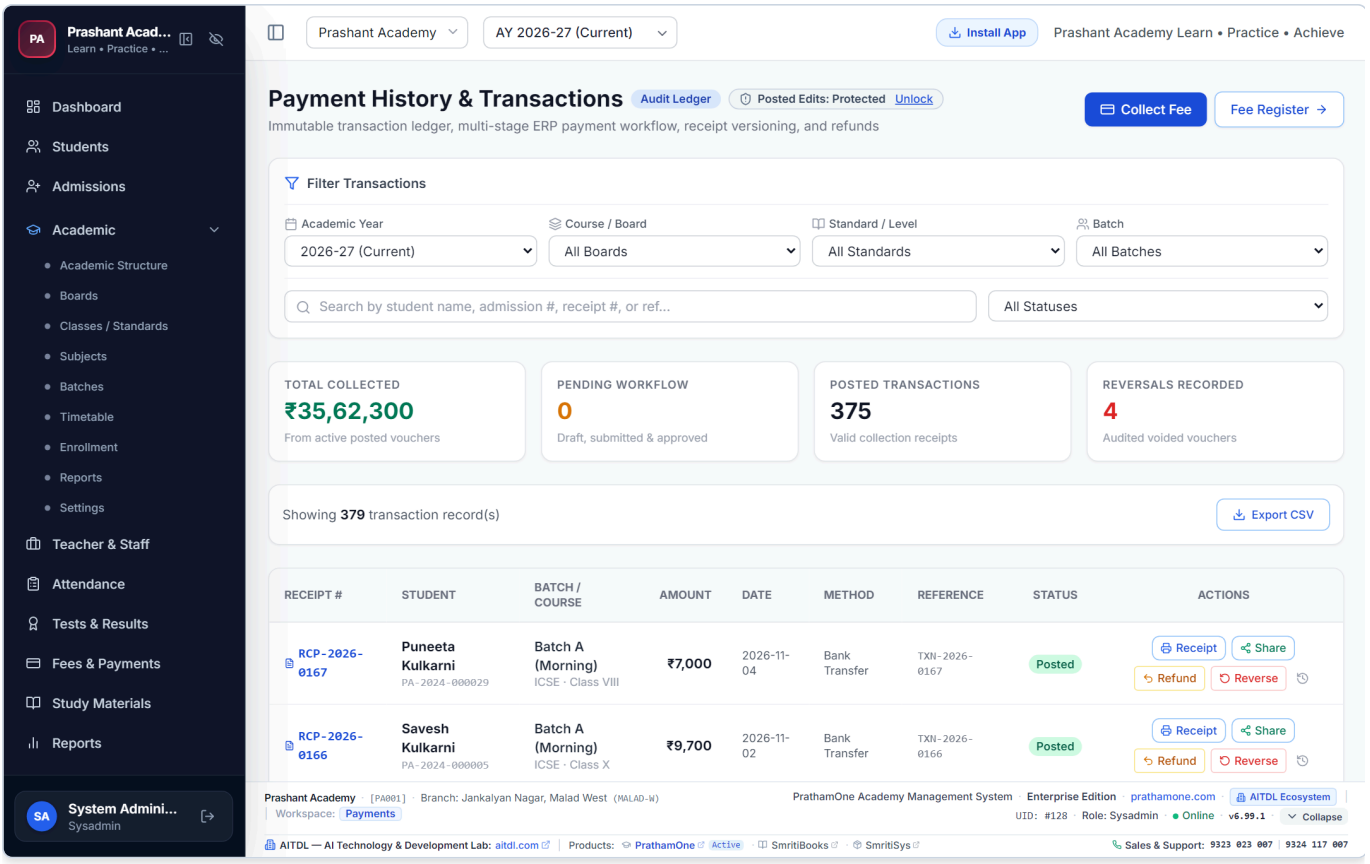


Figure 4.13: Complete Payments Financial Ledger with status filtering and receipt reprint links.

Payment Status Lifecycle:

- `DRAFT`: Initial payment entered by front-desk staff. Editable freely by accountants and staff. No permanent receipt number is locked.
- `SUBMITTED`: Payment submitted for verification. Edits require an explicit administrative reason.
- `POSTED`: Fully verified and posted to the general financial ledger. A permanent, sequential receipt number (e.g., `REC-2026-000142`) is generated and locked.
- `REFUNDED`: Transaction has undergone a full or partial refund. Refund records are maintained as distinct negative ledger entries linked to the original payment via `parent_payment_id`.

> **Financial Control: Posted Payment Security Gate**
>
> Ordinary staff, teachers, and accountants CANNOT edit or delete a POSTED payment. In order to protect ledger integrity and comply with accounting audit standards, only a user with the explicit `payments:edit_posted` permission (Sysadmin) can unlock and edit a posted transaction.

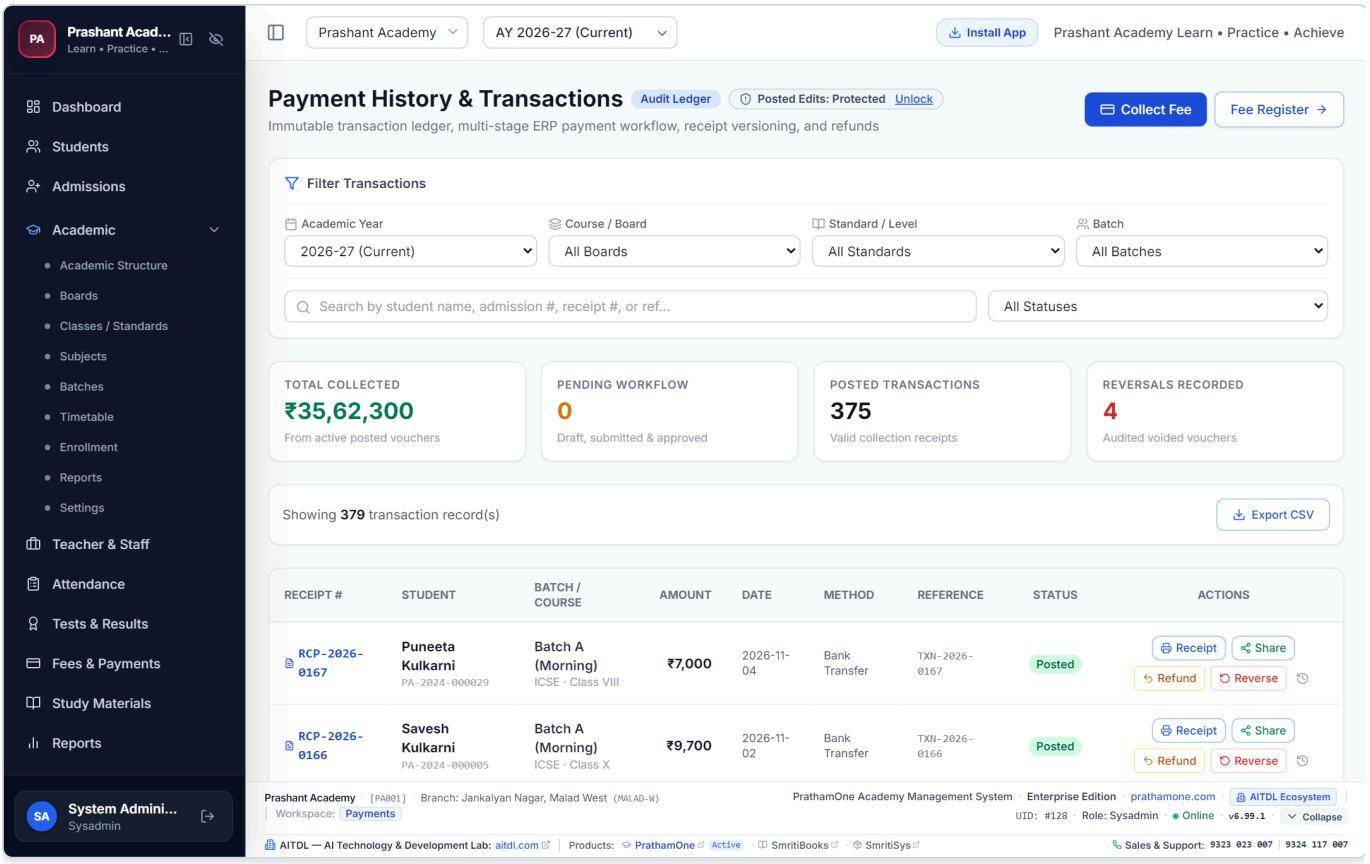


Figure 4.14: Posted Payment Security Gate modal enforcing authorization and mandatory reason logging.

4.9 Import Fee Register (CSV/Excel Wizard)

Route: `/fees/import` | **Access:** Accountant, Admin, SysAdmin

Purpose: Seamless migration of legacy payment books or external bank reconciliation statements into PrathamOne EduOS via structured CSV/Excel upload.

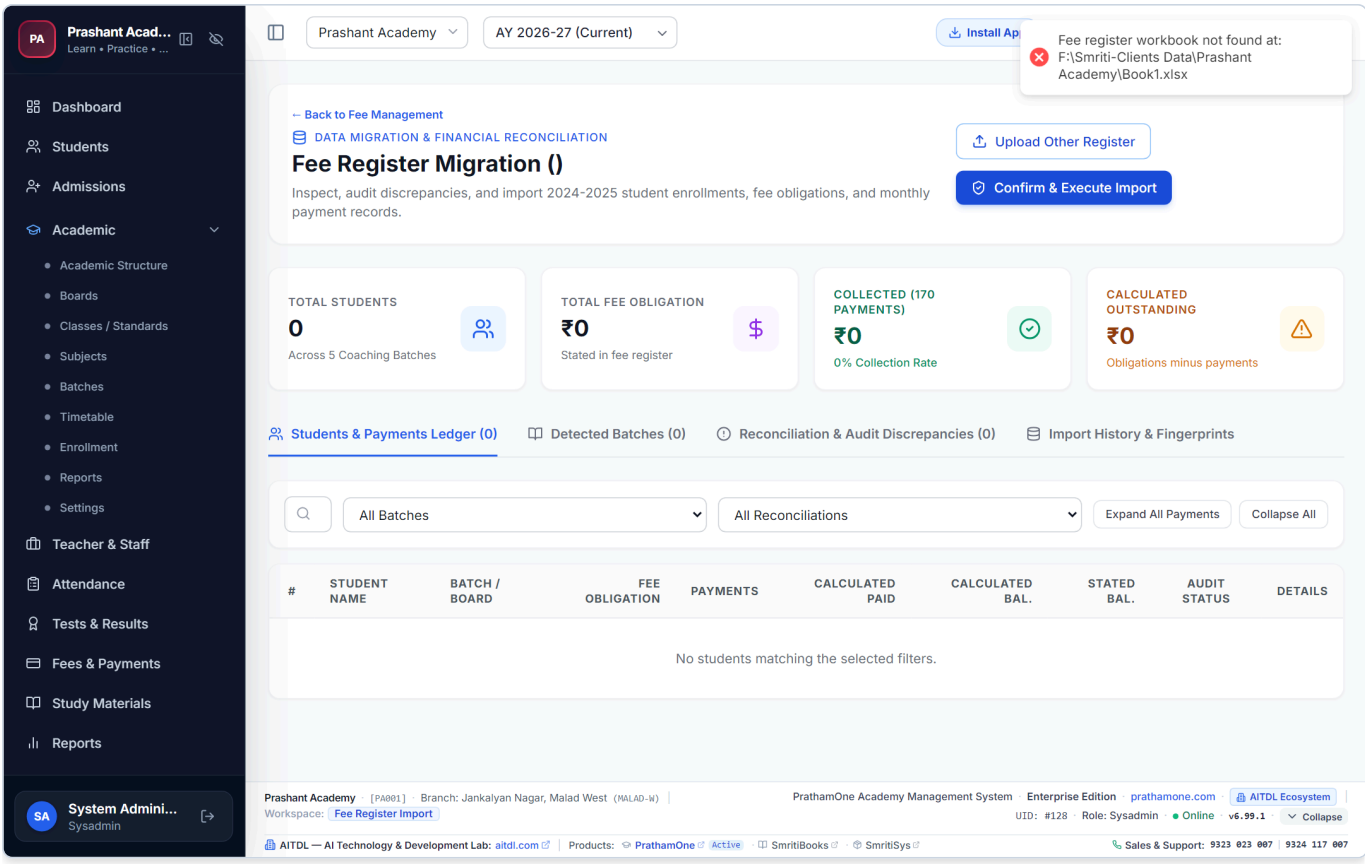


Figure 4.15: Import Fee Register Wizard showing file upload dropzone and column mapping schema.

4.10 Examinations, Grading & Report Cards

Route: `/examinations` | **Access:** Teachers, Admin, Principal, SysAdmin

Purpose: Schedule periodic unit tests, quarterly exams, mock tests, and final examinations. Record subject marks, calculate grade percentages, and print official report cards.

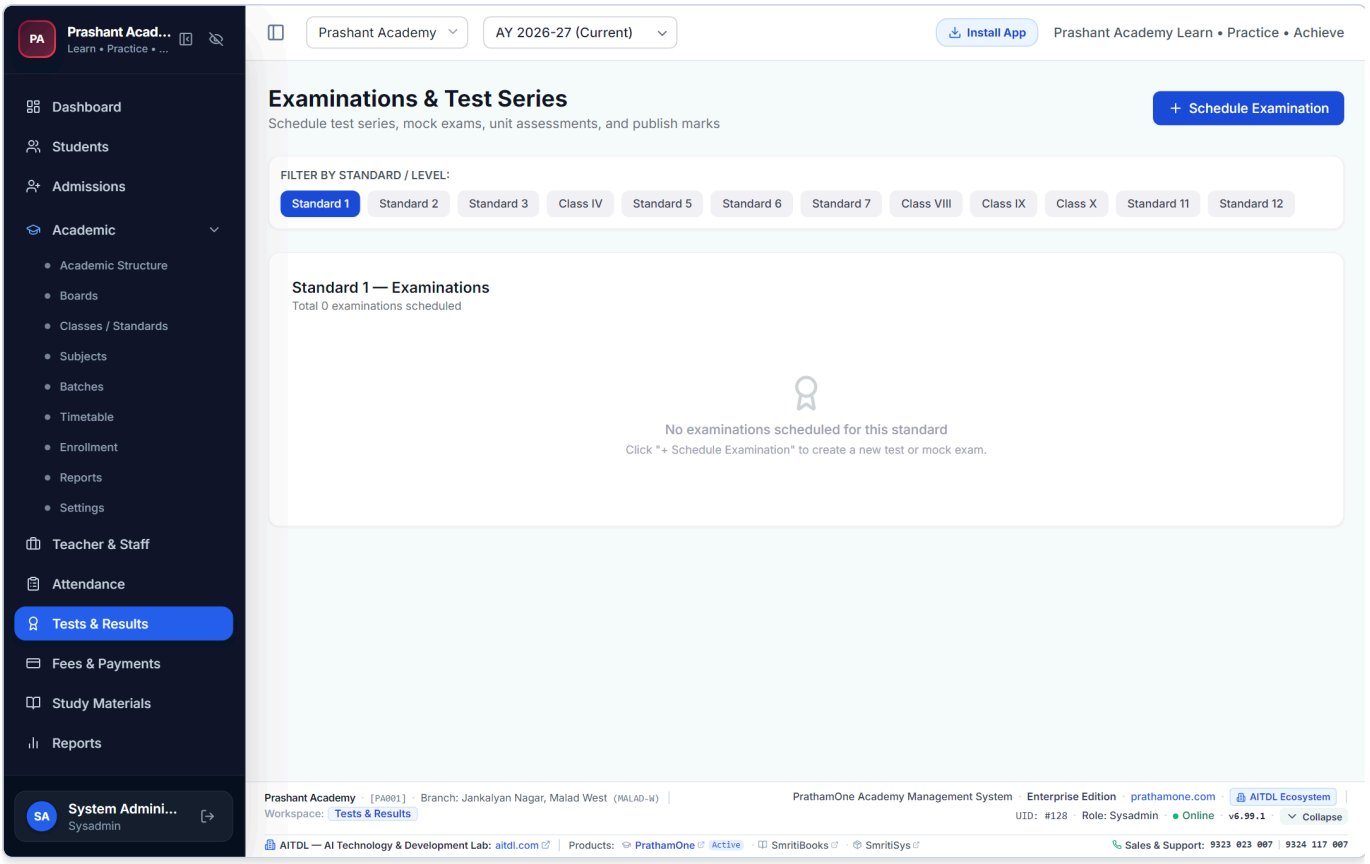


Figure 4.16: Examinations Workspace showing active schedules, marks entry grid, and rank lists.

4.11 Communication Hub & Broadcast Notices

Route: `/communication` | **Access:** Admin, Principal, SysAdmin

Purpose: Institute-wide broadcast messaging for emergency notices, holiday announcements, exam timetables, fee reminders, and parent-teacher meeting circulars.

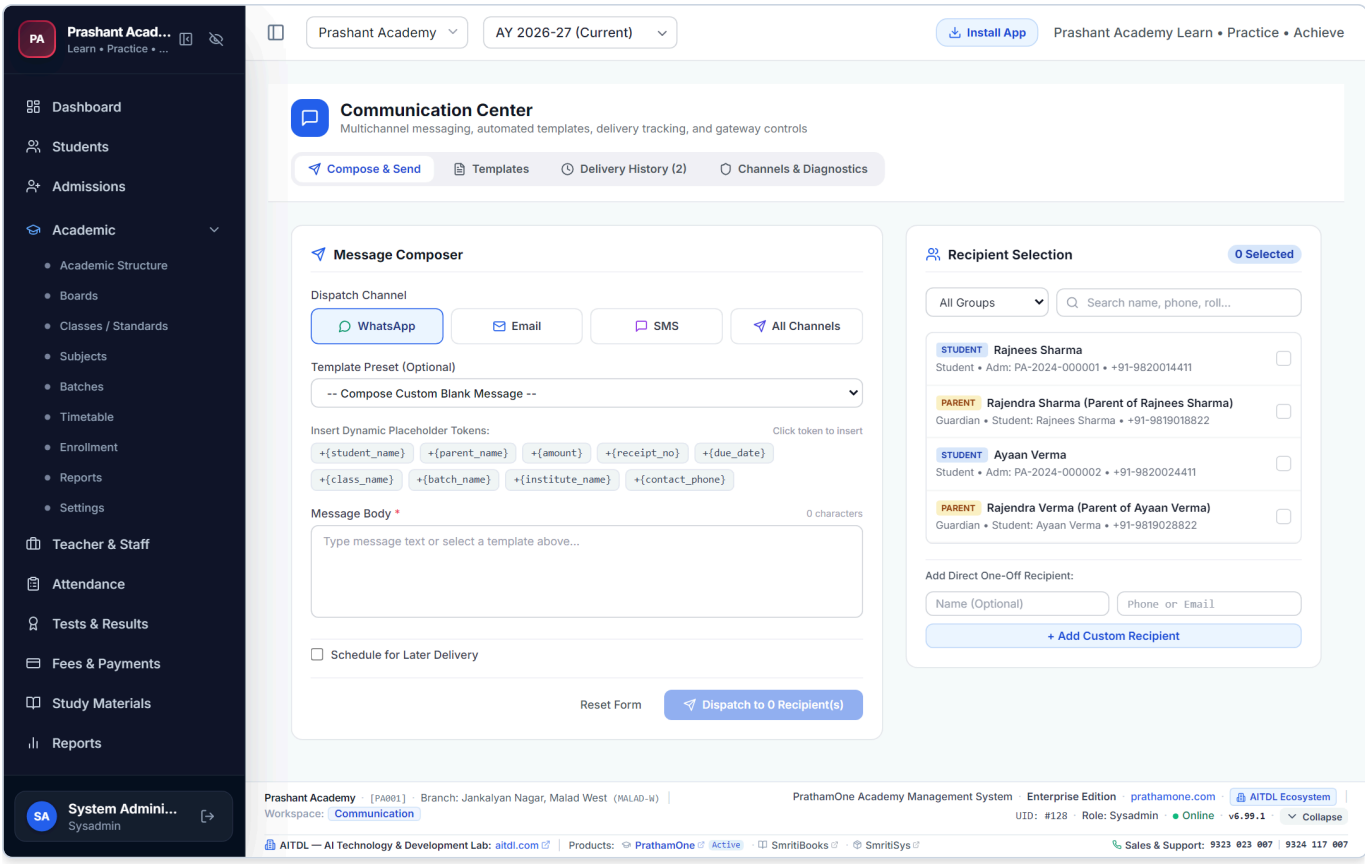


Figure 4.17: Multi-channel Communication Center with audience filters and message logs.

4.12 High-Performance Reports & Analytics Grid

Route: `/reports` | **Access:** Accountant, Admin, Principal, SysAdmin

Purpose: Enterprise-grade tabular reporting engine equipped with real-time column grouping, dynamic pivoting, multi-column sorting, and instant Excel/CSV download.

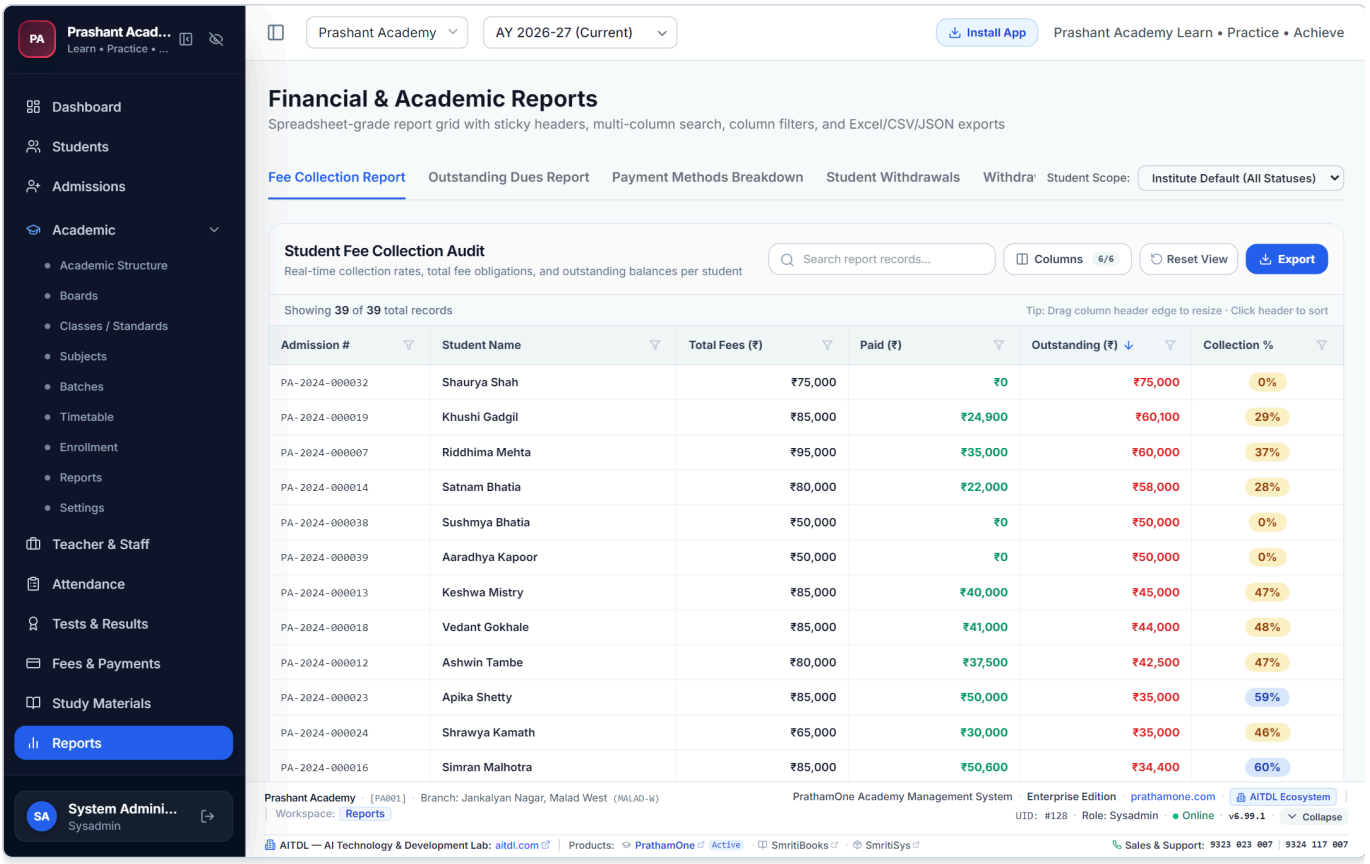


Figure 4.18: High-Performance Excel Report Grid displaying live financial realization analytics.

4.13 Forms & Documents Printing Center (DPDP Compliant)

Route: `/forms` | **Access:** Admin, Principal, Staff, SysAdmin

Purpose: Standardized, print-ready document rendering compliant with statutory educational regulations and Digital Personal Data Protection (DPDP) logging.

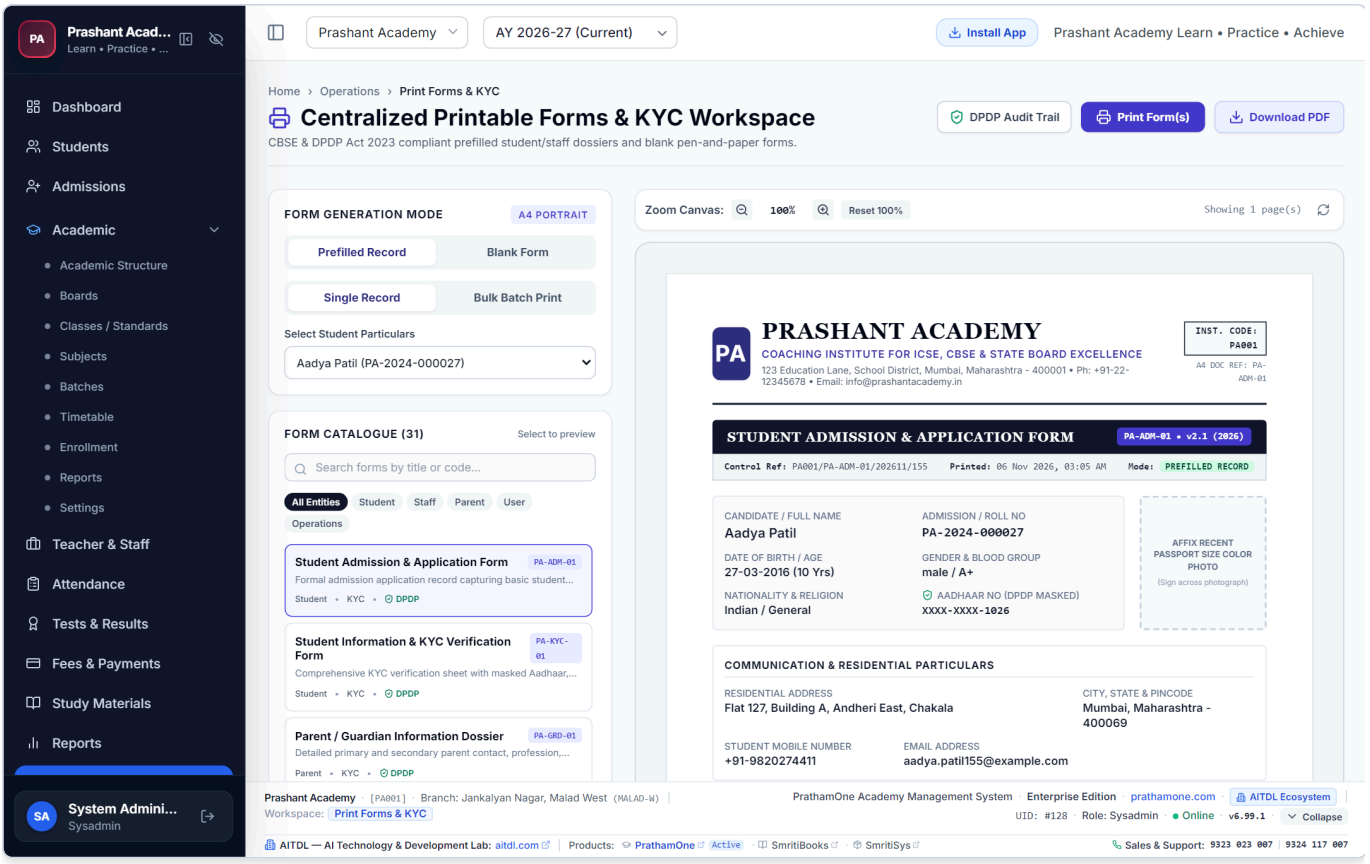


Figure 4.19: Forms & Documents Hub with pre-formatted certificates and DPDP compliance logs.

4.14 Data Import / Export Engine

Route: `/import-export` | **Access:** Admin, SysAdmin

Purpose: Bulk bidirectional synchronization of master data entities via CSV and JSON formats.

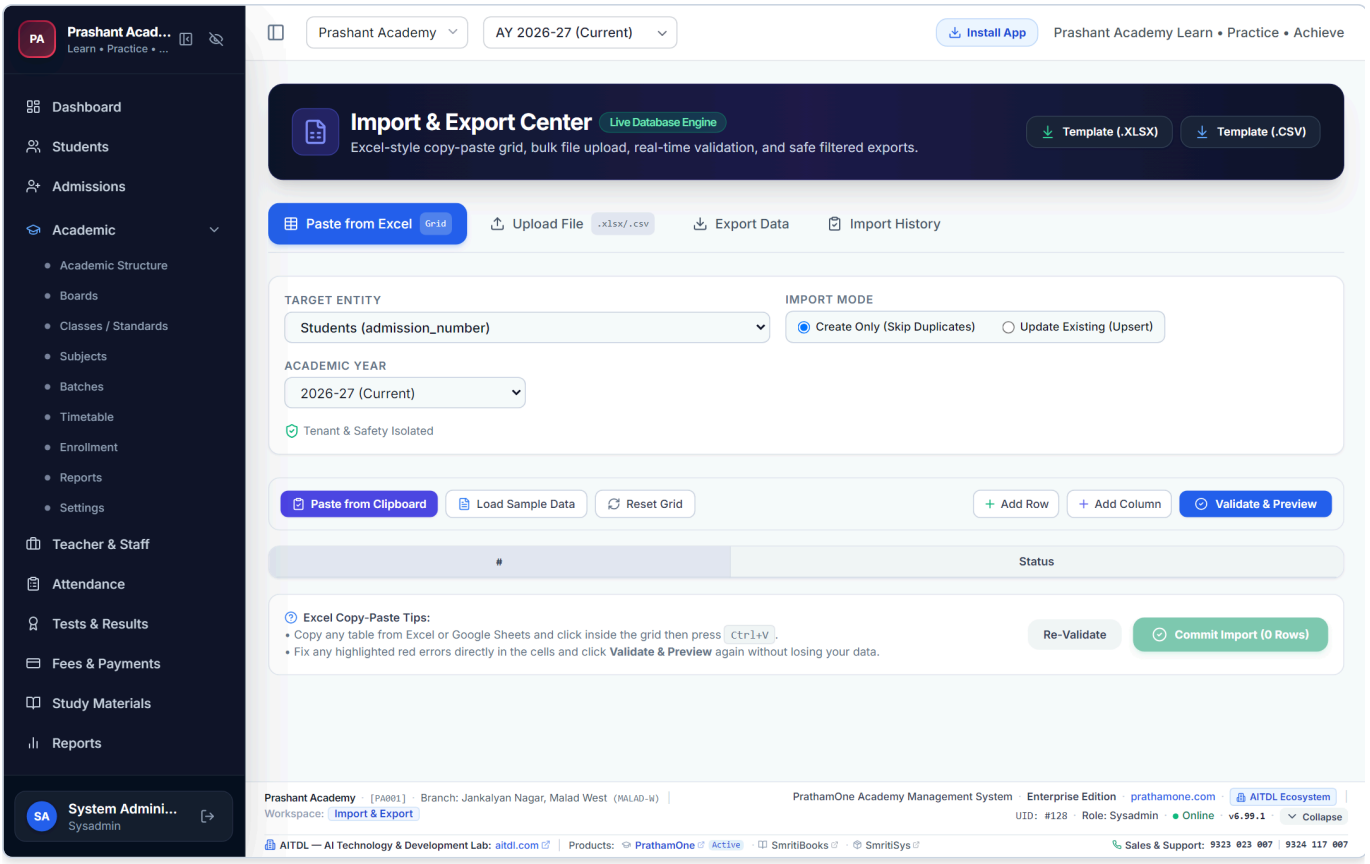


Figure 4.20: Master Data Import and Export Workspace with downloadable CSV templates.

4.15 User Management & RBAC Administration

Routes: `/users`, `/users/{id}` | **Access:** Admin, Superadmin, SysAdmin

Purpose: Provisioning user accounts, assigning roles, resetting credentials, and reviewing permission matrices.

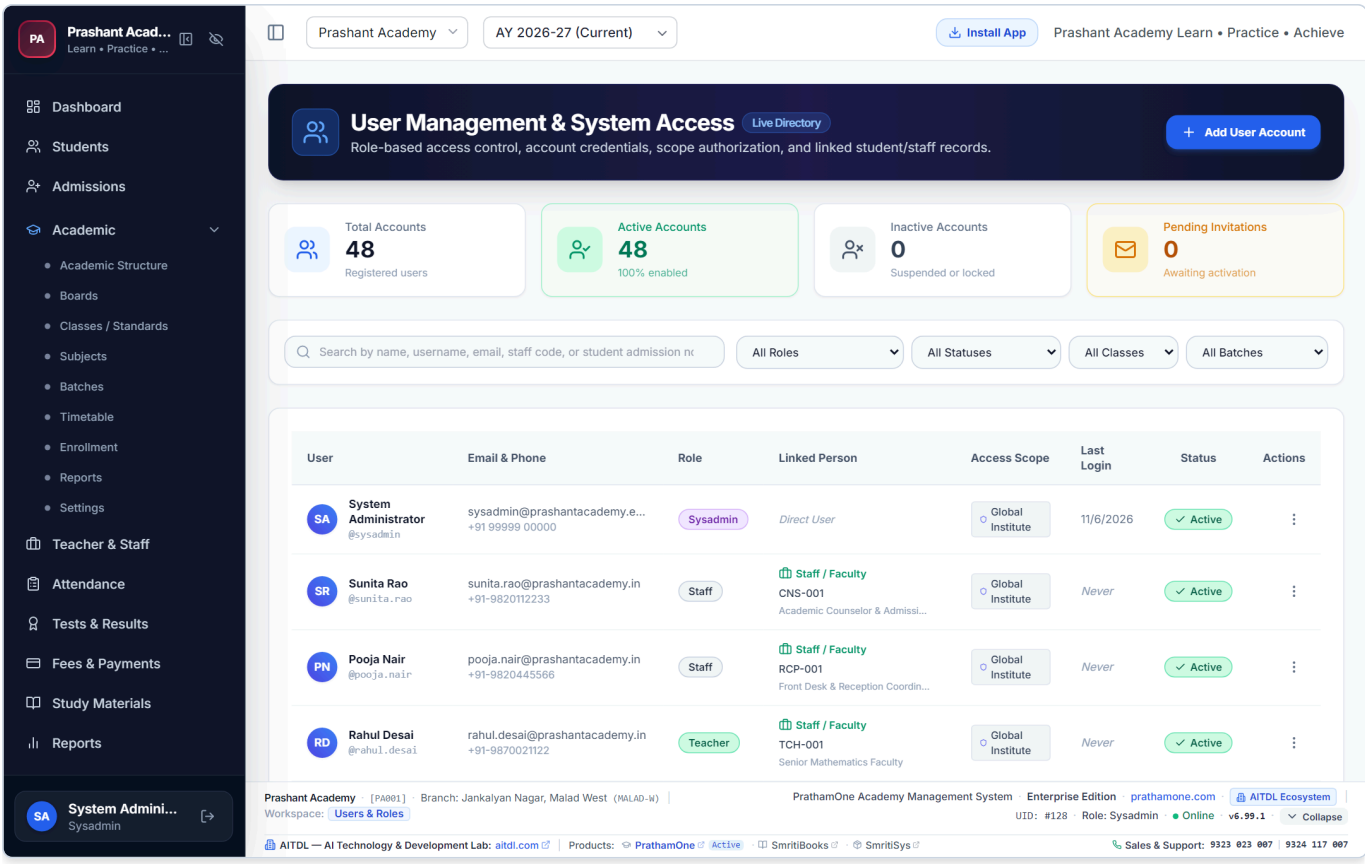


Figure 4.21: User Management Directory displaying user roles, last login timestamps, and status toggles.

4.16 Institute Settings & Live Branding Customizer

Route: `/settings` | **Access:** Admin, SysAdmin

Purpose: Configure institute name, contact info, official logo, and brand color palette with instant real-time preview.

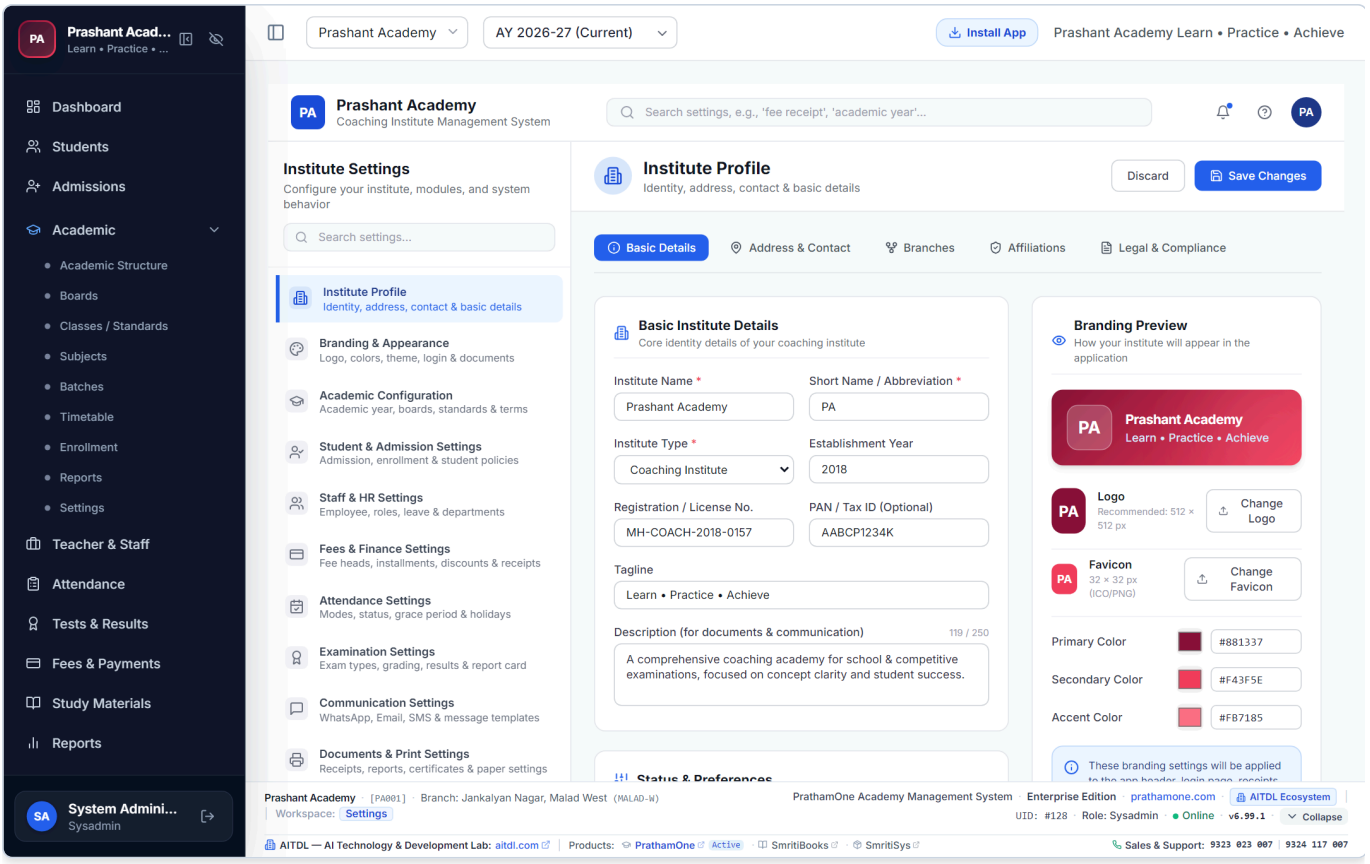


Figure 4.22: Settings Workspace featuring the interactive Live Branding Preview card.

4.17 SysAdmin Center & Universal Records Editor

Route: `/sysadmin` | **Access:** SysAdmin Exclusive

Purpose: Master control center for system diagnostics, server telemetry, immutable security logs, and the Universal Records Editor.

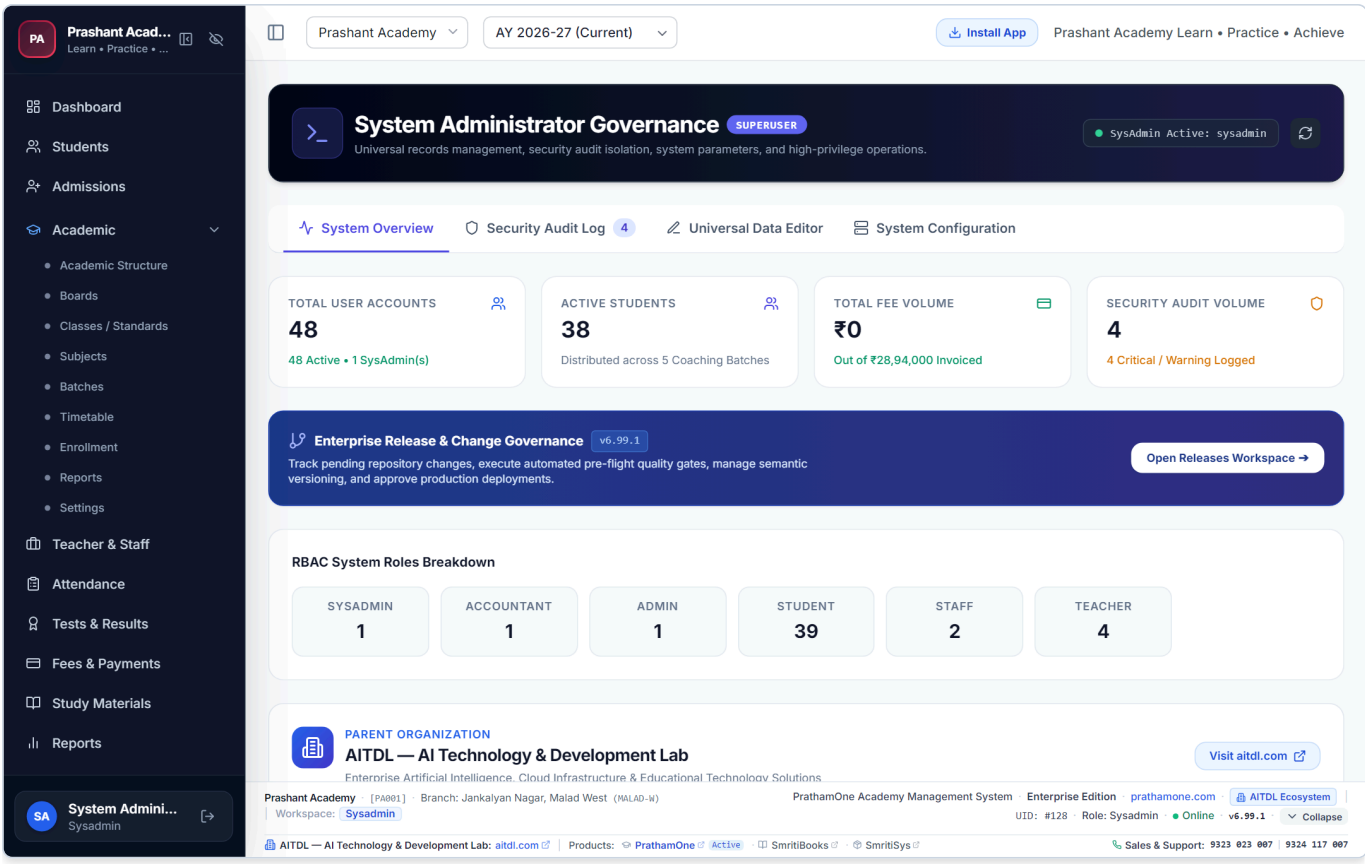


Figure 4.23: SysAdmin Center master dashboard with system health monitors and audit feeds.

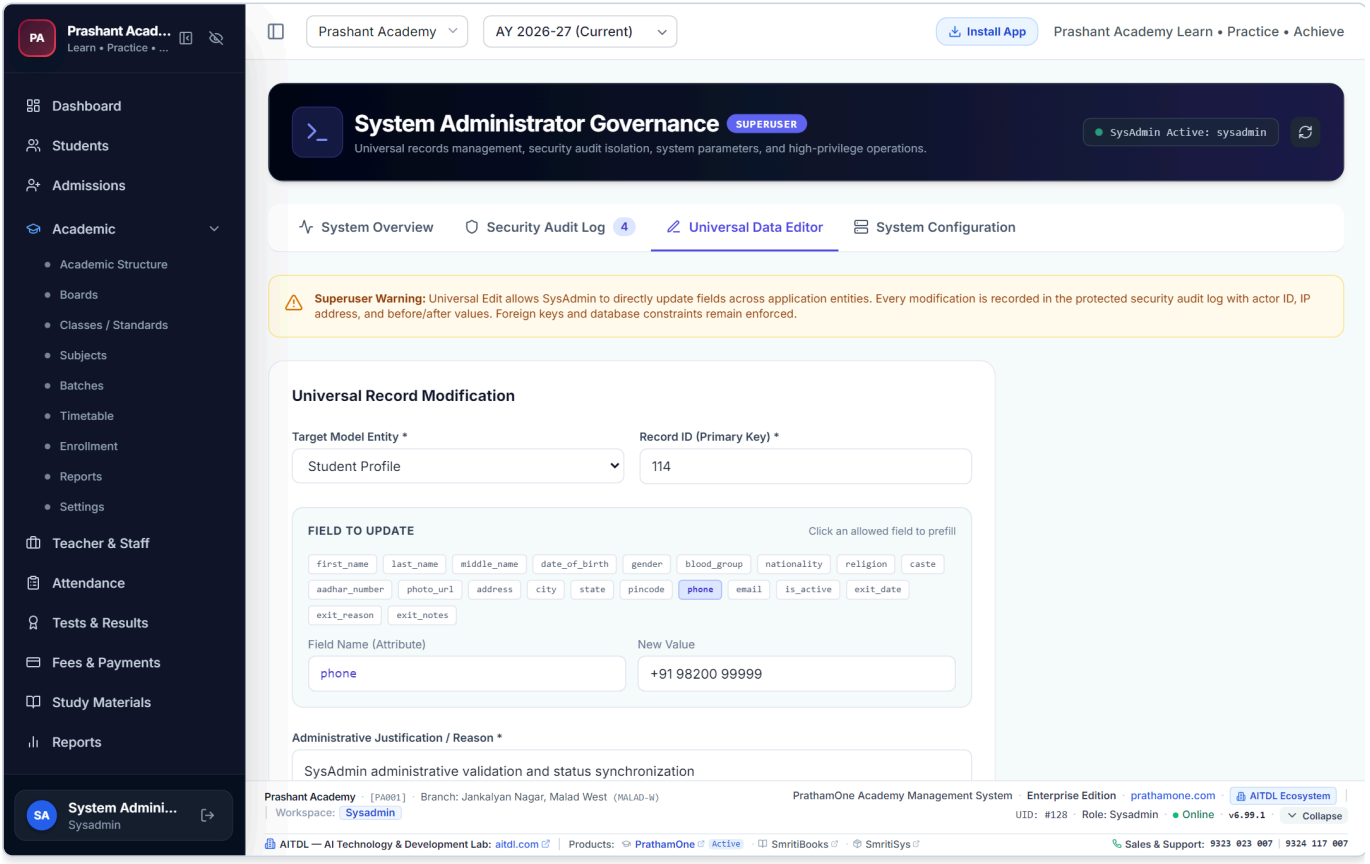


Figure 4.24: Universal Records Editor with clickable permissible field chips and mandatory reason field.

4.18 Release Management & Semantic Version Audit

Route: `/releases` | **Access:** All Users (Read-only)

Purpose: Enterprise release tracking showing current version (`v1.88.2`), build number (`2026.11.12-REV-161-v1.88.2`), release history, and security patches.

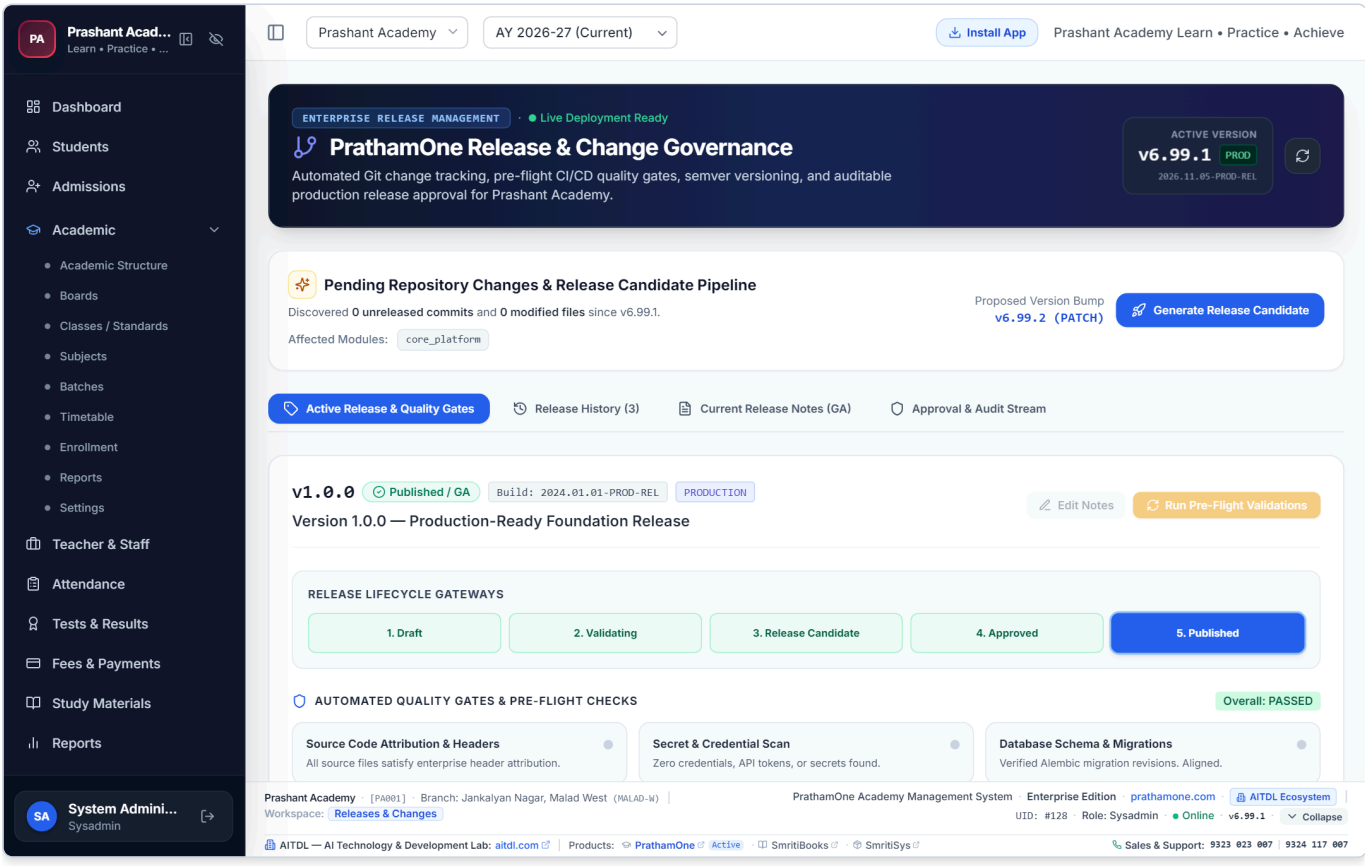


Figure 4.25: Software Release & Changelog Audit workspace.

4.19 CodeX — Configurable Feature Activation & Numeric Coding Studio

Route: `/codex` | **Access:** Business Owner / Authorized SysAdmin Exclusive

Purpose: Optional, highly secure numeric obfuscation engine that masks sensitive business figures (salaries, prices, fees, profit margins) using custom owner-defined alphanumeric symbol mappings.

> **Mandatory System Default Policy: DISABLED (OFF)**
>
> In accordance with global governance directives, CodeX is **DISABLED by default everywhere**. Installing, updating, or initializing software will never auto-activate CodeX. It requires explicit, authorized activation by the Business Owner.

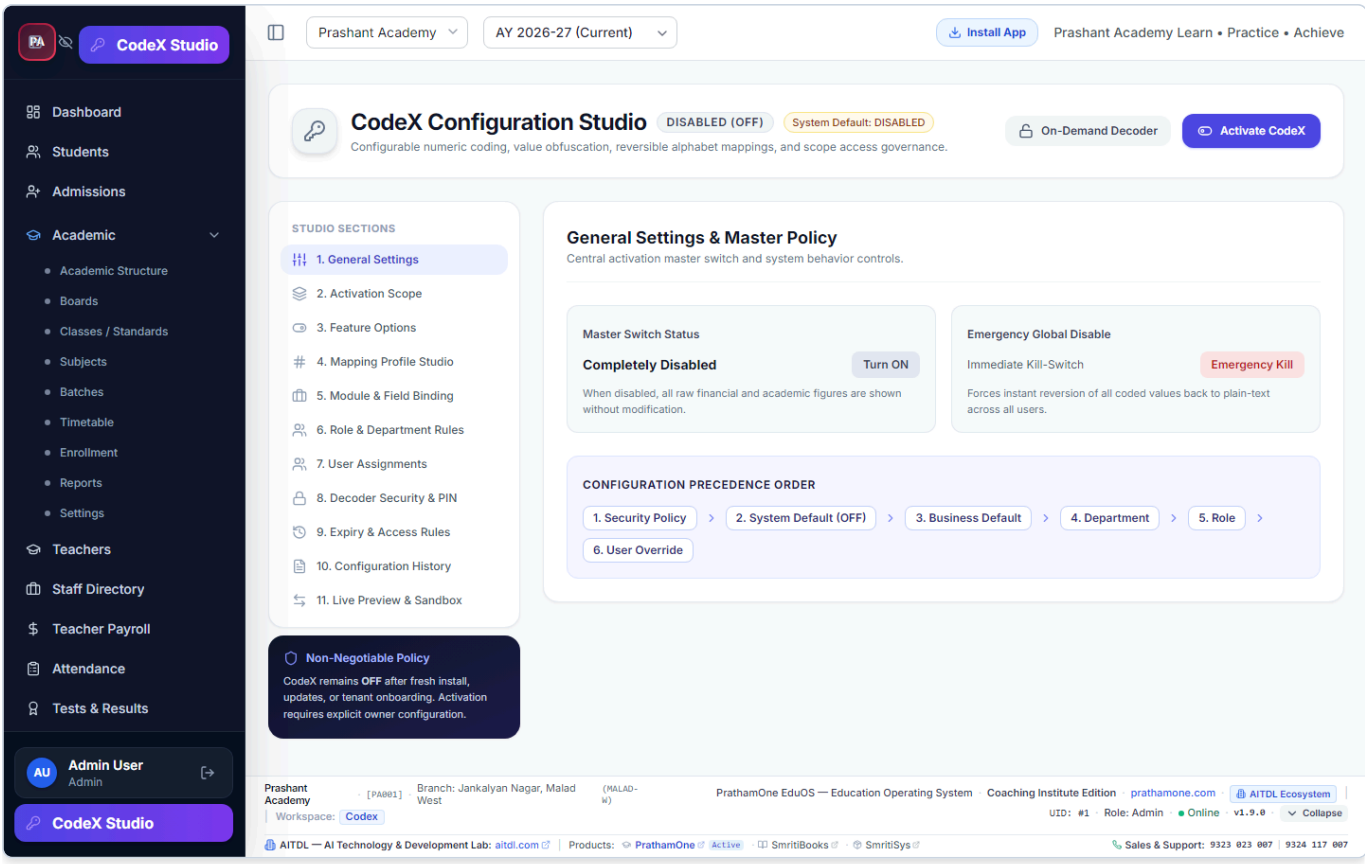


Figure 4.26A: CodeX Configuration Studio featuring master activation toggle and 16 granular feature controls.

Core Architectural Capabilities:

- **Bijective Reversible Mapping:** Digits 0 through 9 map to unique custom symbols (e.g., `0→Q, 1→A, 2→Z, 3→M, 4→K, 5→P, 6→X, 7→B, 8→T, 9→R`). An amount of `₹ 12,500` displays as `AZPQQ`, and decodes back to `12500`.
- **Granular 16-Feature Control:** Independently enable or disable CodeX across Student Fees, Teacher Payroll, Salary Slips, Invoices, Receipts, and Reports.
- **Scoped Visibility Overrides:** Define access policies per Department, Role, or Individual User (e.g., Accountants see real figures while front desk sees coded values).
- **PIN Protection & QR Decoder:** Decoding coded numbers requires administrative PIN verification. Mobile managers can scan QR codes to decode paper receipts.

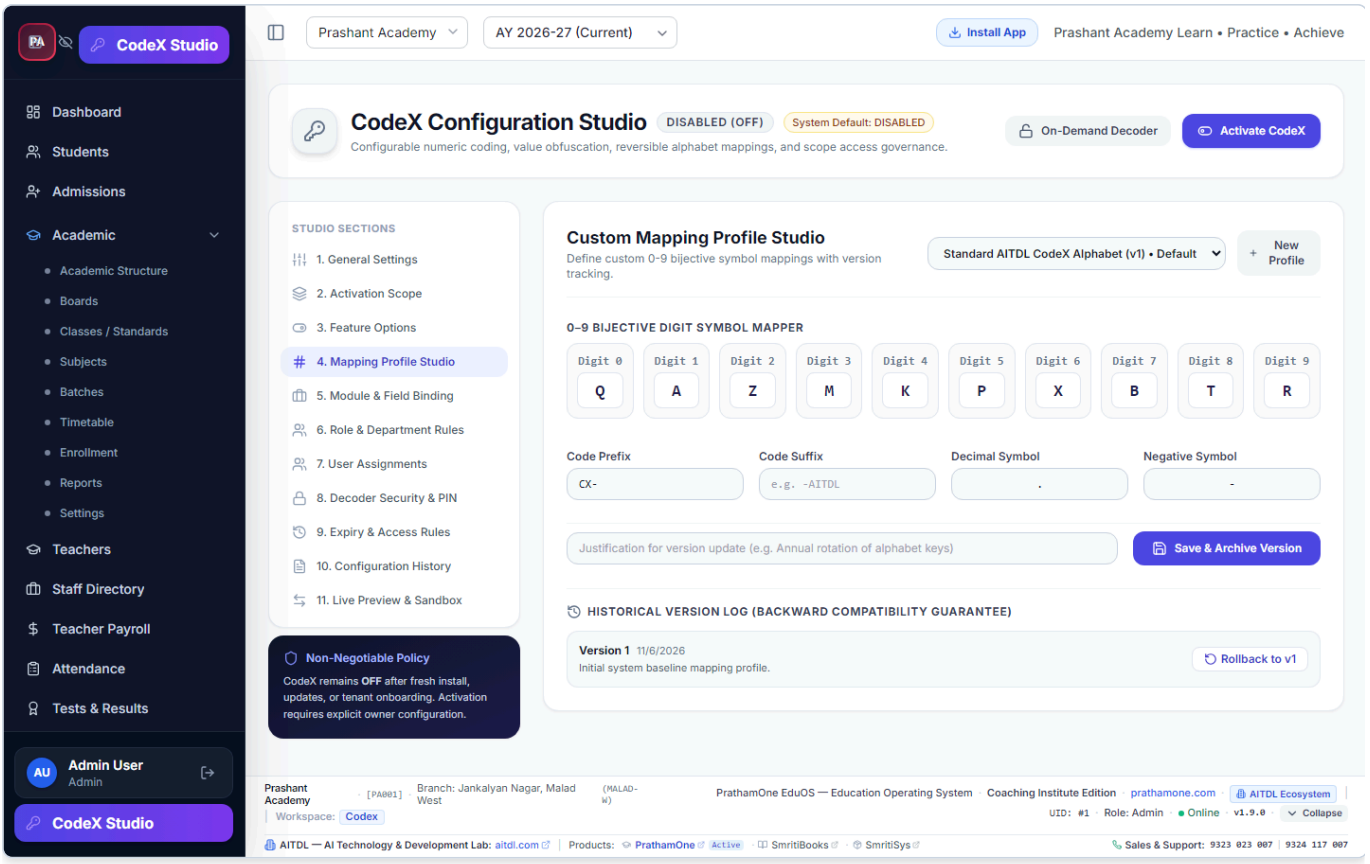


Figure 4.26B: Interactive Digit Mapping Studio with live substitution preview.

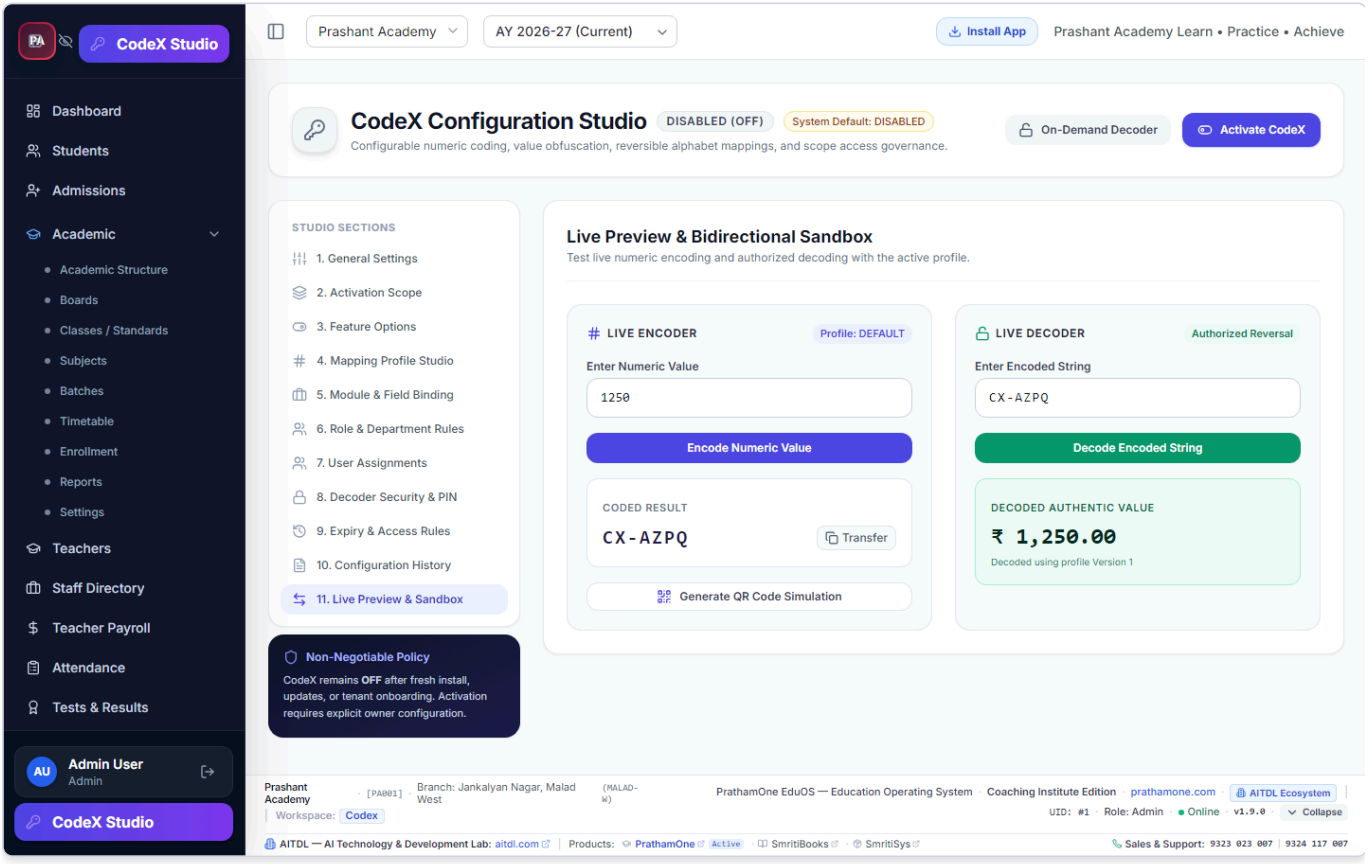


Figure 4.26C: Verification Sandbox testing live bidirectional encoding and decoding before deployment.

4.20 Universal Print Fit & Page Layout Engine

Route: Integrated across all document print triggers | **Access:** All Roles

Purpose: Standardized print layout engine delivering pixel-perfect document rendering across A4, A5, A3, Letter, Legal, and custom paper sizes with zero clipping.

Universal Print Highlights

- **Dynamic Sizing:** Supports standard office paper (A4, Letter) as well as compact receipt formats (A5, Thermal Rolls).
- **Automatic Scaling:** Eliminates page overflow issues by auto-calculating CSS scale factors to fit wide tables onto a single sheet.
- **Thermal Printer Compatibility:** Automatically formats fee receipts for POS thermal printers (80mm/58mm) without manual re-configuration.

4.21 Homework & Assignment Studio (Phases 1–15)

Route: `/homework` | **Student View:** `/student-portal` | **Access:** Teachers, Academic Coordinators, Admin, Students

Purpose: End-to-end homework lifecycle supporting Batch-level, Individual Student-level, and Selected Group targeting, rich attachments, student file/text submissions, late-joiner backfill, batch-transfer carryover, and teacher evaluation with live analytics.

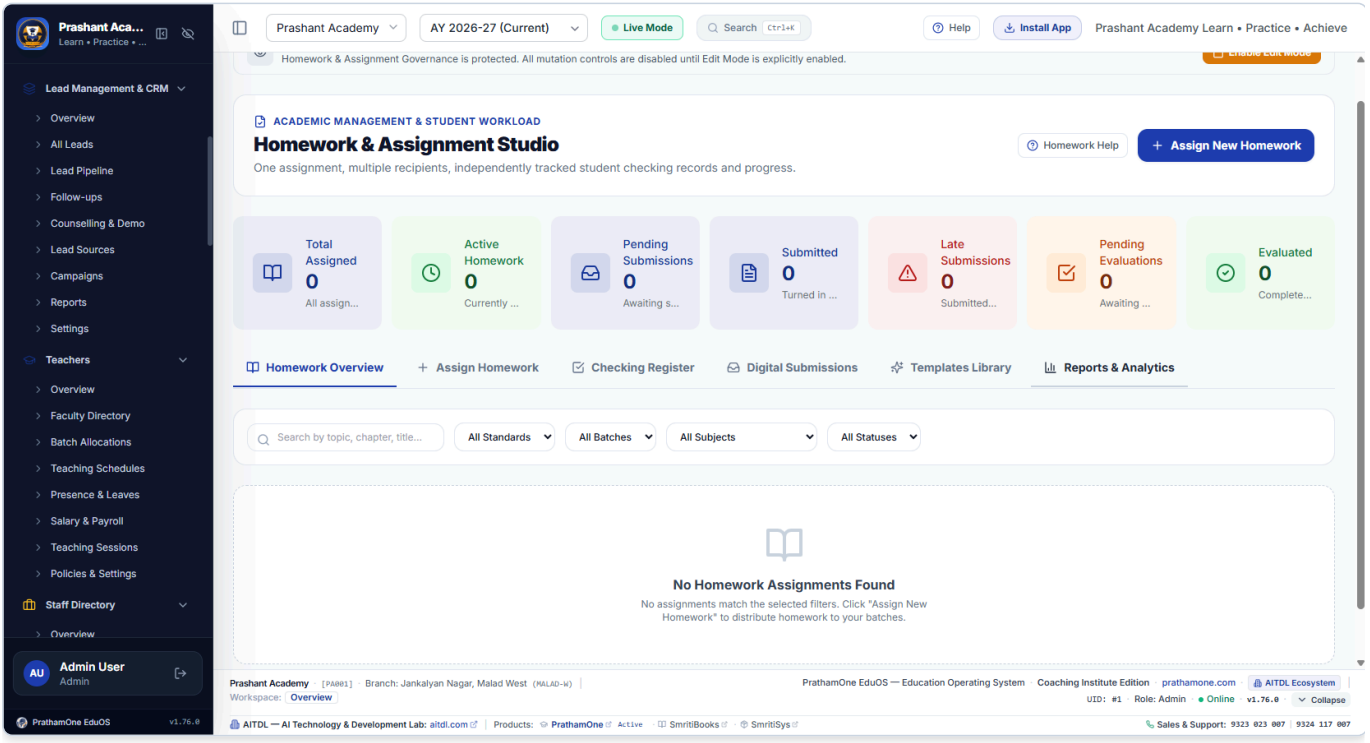


Figure 4.21a: Homework & Assignment Studio workspace with live completion metrics.

Creating & Targeting Assignments

- **Three Assignment Scopes:** Assign work to an entire **Batch** (with frozen recipient snapshot), a curated **Selected Subset**, or **Individual Students** with custom per-student deadlines or exemptions.
- **Draft → Publish Workflow:** Teachers save assignments in Draft state, attach PDFs or worksheets, and click **Publish** to notify targeted students and parents immediately.
- **Lifecycle Automation:** Newly enrolled batch joiners automatically inherit active batch assignments, while batch transfers preserve submission history seamlessly.

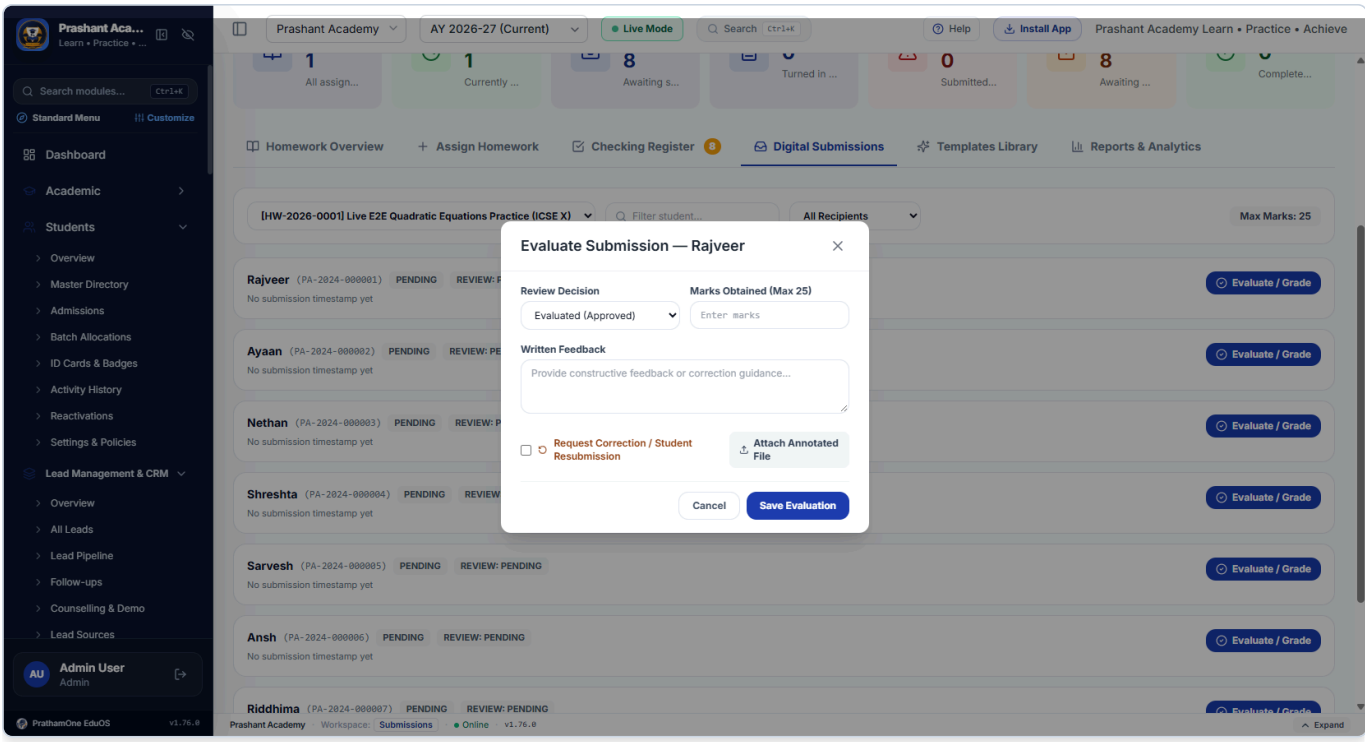


Figure 4.21b: Submission review drawer with marks, remarks, and resubmission controls.

4.22 PrathamOne EduBoard Interactive Classroom

Route: `/eduboard` | **Student Archive:** `/student-portal` | **Access:** Teachers, Coordinators, Admin, Students (Read-only published boards)

Purpose: Built-in multi-page digital classroom engine integrated directly with Academic Batches, Subjects, Teaching Sessions, Timetables, and Student Portal revision archives.

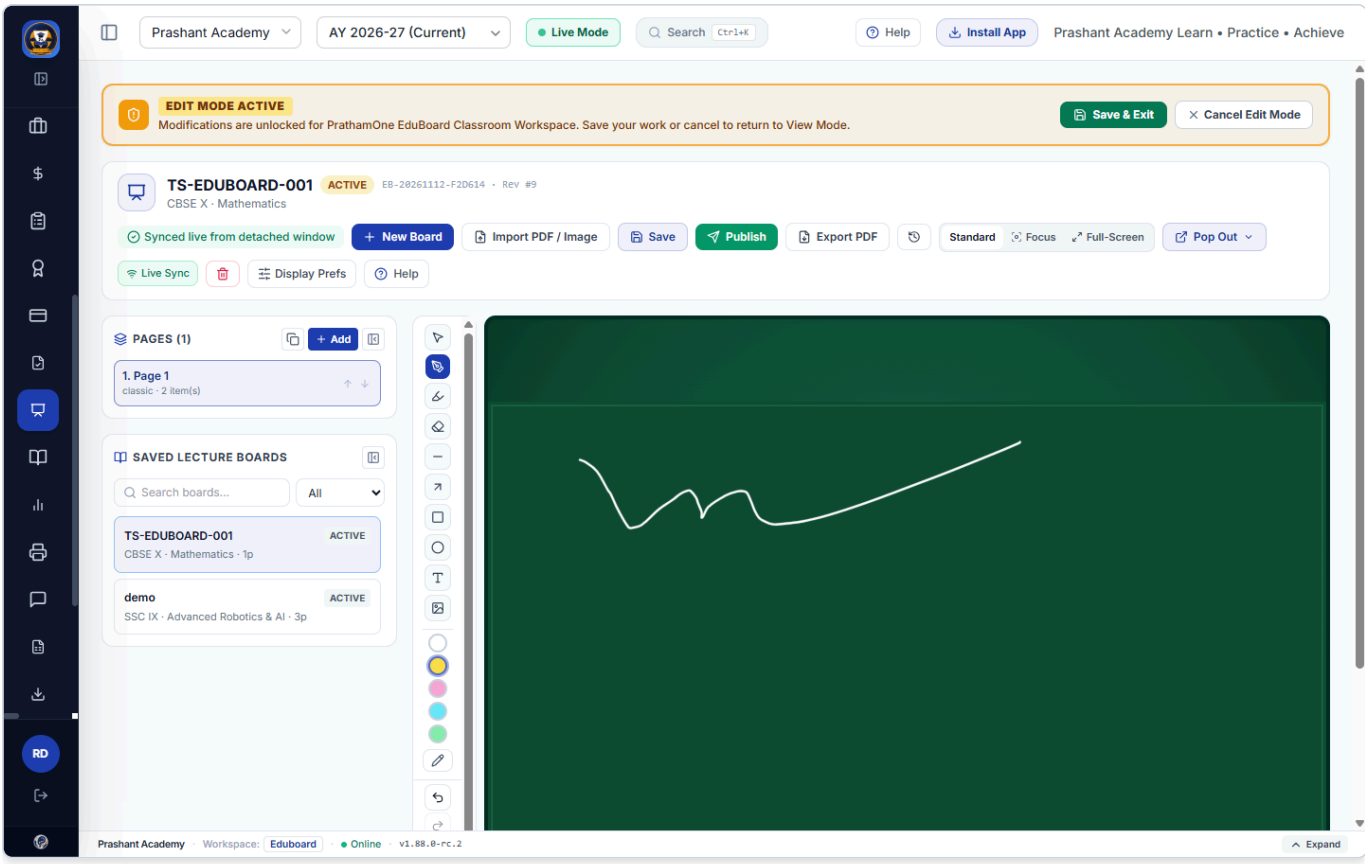


Figure 4.22a: PrathamOne EduBoard running in Classic Greenboard theme with collapsible Pages & Saved Boards drawers.

Workspace Display Modes & Themes

- **Three Display Modes:** Switch seamlessly between **Standard Mode** (with independently collapsible Pages and Saved Lecture Boards panels), **Focus Mode** (maximized canvas with drawer overlays), and **Immersive Fullscreen Mode** (native browser Fullscreen API + viewport fallback and Escape-key exit).
- **Board Themes:** Choose **Whiteboard**, **Whiteboard Grid**, **Classic Greenboard**, or **Blackboard** with high-contrast chalk/marker palettes (White, Yellow Chalk, Sky Blue, Rose, Mint).
- **Canvas & Teaching Tools:** Vector Pen, Highlighter, Eraser, Line, Arrow, Rectangle, Circle, inline Text, Undo/Redo, zoom/pan, multi-page reordering, and multi-page PDF lecture import.
- **Dual-Screen Pop-Out Sync:** Teachers can detach a live projector/external monitor window using **Pop-Out Board** powered by cross-tab `BroadcastChannel` + database synchronization, then click **Publish** or **Export PDF** to share lecture notes directly to Student Portal.

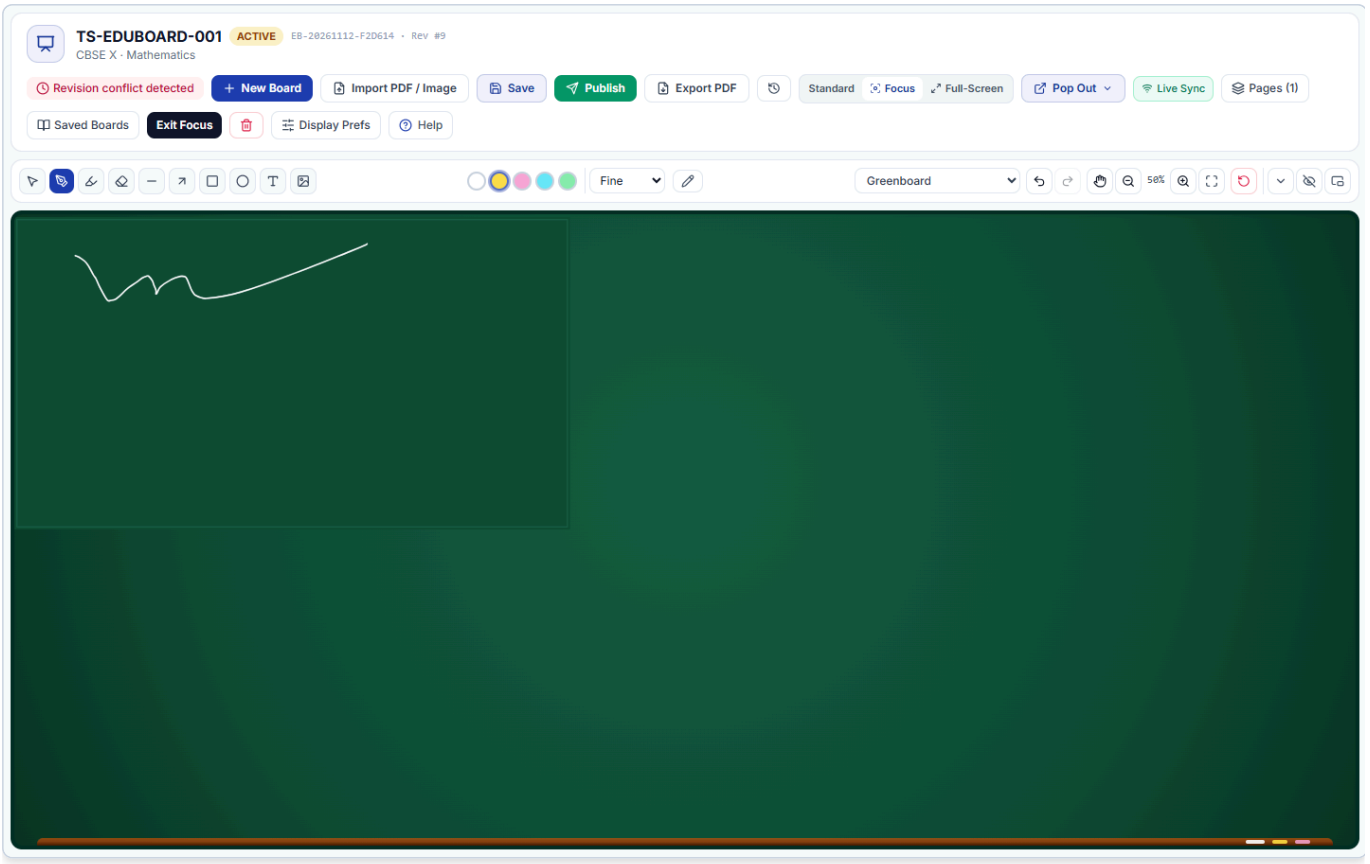


Figure 4.22b: Focus Mode maximizing classroom canvas real estate with compact floating tools.

4.23 Official Fee Receipt Printing & PDF Parity (A5 Compact & A4 Portrait)

Route: `/fees` & `/payments` → **Payment Receipt View**

Purpose: Guarantees 1:1 visual parity between the on-screen Receipt Preview, browser Print Preview (`Ctrl+P` / **Print Receipt**), and **Download PDF** output across both **A5 Compact** and **A4 Portrait** paper formats.

- **Full Institute Branding Preserved:** Academy crest logo, institute name, full street address, helpline phone numbers, email address, QR verification code, installment ledger summary, and authorized signature render identically in Print Preview and exported PDFs.
- **Isolated Print Frame Architecture:** Clicking **Print Receipt** activates a dedicated print container sized cleanly to the selected A5 or A4 geometry with crisp color retention (`print-color-adjust: exact`) and zero unwanted sidebar/header cropping or shrinking.

4.24 Help & About Centre, Documentation Sync & Governance

Route: `/help` | **Shortcut:** Top Navbar **Help** button | **Access:** All Authenticated Users

Purpose: Centralized in-app knowledge base covering **20 platform modules**, interactive tutorials, troubleshooting guides, printable SOPs, Support Ticket Desk, Release Changelog (synced with live backend version metadata), and Third-Party Open-Source Acknowledgements.

- **Verified v1.88.2 Guides:** Includes dedicated guides for *Homework & Assignment Studio*, *PrathamOne EduBoard Interactive Classroom*, *A5/A4 Receipt Printing & PDF Export Parity*, and updated open-source dependency disclosures (PyMuPDF, Pillow, Konva/Canvas, React 18, FastAPI, PostgreSQL 16).
- **Strict Data Privacy & RBAC:** Technical System Diagnostics and Infrastructure telemetry remain restricted strictly to authorized Sysadmin accounts, while Help articles are auto-synchronized on startup without exposing secrets or stack credentials.

5. End-to-End Business Workflows

5.1 Workflow 1: Student Onboarding from Admission to Batch Enrollment

1. **Inquiry & Registration**
   Parent/Student arrives. Front desk opens `/students/add` and enters personal, contact, and academic details.

2. **Batch & Division Assignment**
   Student is assigned to the appropriate Standard (e.g., 10th CBSE) and Batch (Morning Champions Batch A). Roll number is allocated.

3. **Fee Plan Commitment**
   The standard fee plan is automatically mapped to the student profile, establishing the student's fee obligation and installment timeline.

4. **Identity Card & Admission Form Generation**
   One-click generation of the Student ID Card and Admission Record from `/forms`.

5.2 Workflow 2: Financial Collection, Receipting, Correction & Refunds

1. **Payment Collection**
   Accountant receives cash, cheque, or UPI payment via `/payments` → *Record Payment*.

2. **Sequential Receipt Issuance**
   System generates permanent sequential receipt (`REC-2026-XXXXXX`) and posts to financial ledger.

3. **Posted Payment Correction (If Required)**
   If a mistake occurred, Sysadmin accesses the Posted Payment Security Gate with an explicit reason. The original receipt is marked as superseded, and a new versioned receipt is issued. Historical numbers are never deleted.

4. **Processing a Refund**
   If a student withdraws, an authorized accountant processes a linked refund via the Payments module. The refund references the original payment ID and updates the net balance.

6. Special SysAdmin Security Manual

6.1 Universal Record Editor Guardrails & Allowlists

The **Universal Records Editor** (available exclusively to SysAdmin at `/sysadmin`) allows surgical data correction across 8 core database models without direct database manipulation. To prevent relational corruption or security bypasses, strict field allowlists are enforced:

Target Model	Permissible Editable Fields	Strictly Prohibited Fields
`student`	`first_name, last_name, middle_name, date_of_birth, gender, blood_group, nationality, religion, caste, aadhar_number, photo_url, address, city, state, pincode, phone, email, is_active, exit_date, exit_reason, exit_notes`	`id, institution_id, registration_number, created_at, updated_at`
`staff`	`first_name, last_name, designation, department, qualification, experience_years, phone, email, address, status, joining_date, specialization`	`id, institution_id, staff_code, created_at, updated_at`
`user`	`first_name, last_name, email, phone, role, is_active`	`id, institution_id, username, hashed_password, created_at`
`fee_structure`	`name, description, total_amount, is_active`	`id, institution_id, standard_id, created_at`
`division / batch`	`name, code, room_number, capacity, schedule_timing, status, is_active, class_teacher_id`	`id, institution_id, standard_id, created_at`
`standard`	`name, level, description, is_active`	`id, institution_id, board_id, created_at`
`institution`	`name, tagline, address, city, state, pincode, phone, email, website, logo_url, favicon_url, primary_color, secondary_color, accent_color, description, is_active`	`id, code, created_at`

SysAdmin Anti-Lockout Safeguard

The backend actively prevents the primary SysAdmin user from deactivating their own account (User ID 128) via the Universal Editor. Any attempt to set `is_active = false` on the current SysAdmin returns **HTTP 400 Bad Request**.

7. Troubleshooting, Error Codes & Support

7.1 Common Runtime Errors & Resolving Taxonomy Codes

PrathamOne EduOS features a centralized, taxonomy-based error monitoring architecture. When an error occurs, a user-friendly modal is displayed along with a unique reference code (e.g., `ERR-20261106-XXXX`):

Taxonomy Category	Typical Root Cause	Recommended User Resolution
INTERNET_DISCONNECTED	Local Wi-Fi or ethernet drop.	Check router connection. The application PWA will operate in offline cache mode until connection is re-established.
SESSION_EXPIRED (HTTP 401)	JWT token lifetime exceeded (8 hours).	The system automatically redirects you to `/login` once. Re-enter your password to renew session.
PERMISSION_DENIED (HTTP 403)	Current user role lacks permission for the requested action.	Contact your institute Administrator or Sysadmin to assign appropriate permissions.
RESOURCE_NOT_FOUND (HTTP 404)	Target student ID or record does not exist or was removed.	Verify the record ID in the directory. Deleted records can be inspected in the Audit Log.
VALIDATION_FAILED (HTTP 400/422)	Missing mandatory field (e.g., Edit Reason) or duplicate unique key.	Review highlighted input fields in red. Ensure an administrative reason is provided for transaction modifications.

7.2 Frequently Asked Questions (FAQ)

Q: Can I edit a fee payment after the receipt has been printed?
A: Yes, but only with authorized Sysadmin credentials via the Posted Payment Security Gate. The system preserves the original receipt for audit compliance and marks it as superseded.

Q: How do I export my student roster to Excel?
A: Go to **Students** or **Reports**, set your desired filters, and click the green **Export to Excel / CSV** button at the top right of the table.

Q: What happens when a student withdraws? Does it delete their fee history?
A: No! Financial records and attendance history are permanently preserved. The student's status is simply marked as Exited, freeing up batch capacity.

7.3 Official Support Channels & SLA

PrathamOne EduOS Official Support Helpdesk

Direct Telephone / WhatsApp Support:
- Primary Helpline: **+91 9323 023 007**
- Secondary Helpline: **+91 9324 117 007**

Official Support Email:
- Primary Desk: support@prathamone.com
- Alternate Desk: prathamone@outlook.com

Official Product Portal:
- https://prathamone.com

Author & Chief Architect: Jawahar R Mallah
Organization: AITDL — AI Technology & Development Lab